

Cross-factor analysis of software engineering practices versus practitioner demographics: An exploratory study in Turkey



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ARTICLE INFO

Article history:

Received 22 May 2015

Revised 8 September 2015

Accepted 9 September 2015

Available online 24 September 2015

Keywords:

Cross-factor analysis

Software engineering practices

Practitioner demographics

ABSTRACT

Context: Understanding the types of software engineering practices and techniques used in the industry is important. There is a wide spectrum in terms of the types and maturity of software engineering practices conducted in each software team and company. Furthermore, it is important to understand the cross-factor relationship of software engineering practices and practitioner demographics including their companies and projects, e.g., is it the case that practitioners developing software for sectors such as military would utilize software size measurement approaches more, or use different software development methodologies, compared to practitioners developing software for other sectors?, and what kinds of practices are conducted by practitioners in small versus large companies?

Objective: Our objective is to get an understanding into the cross-factor correlation of various software engineering practices versus practitioner demographics including their companies and projects (e.g., target industry, size and work experience). Such an understanding will enable us to identify patterns and pinpoint special issues which should be studied and addressed in the context of each specific demographic (e.g., small versus large companies). Thus, we decided to conduct an exploratory study in this area and collected real industrial data in the context of Turkey which has a vibrant software industry.

Method: To achieve the above objective, we use the data from a recent Turkish-wide survey of software engineering practices which was systematically designed with 46 questions based on our past experience in the Canadian context and using the Software Engineering Body of Knowledge (SWEBOK). 202 practicing software engineers participated in the survey. We raise a set of 12 research questions about the cross-factor correlation of software engineering practices and practitioner demographics, and address them using statistical analysis.

Results: The exploratory study results reveal important and interesting findings about cross-factor relationship of software engineering practices and practitioner demographics. Among some of the most interesting findings are the followings: (1) By analyzing the trends, we were first surprised to see that as a practitioner gets more years of work experience, against what one would expect, s/he experiences more challenges in Software Development Life-Cycle (SDLC) phases and SE tasks; (2) Almost 55% of participants measure software size; (3) Agile/lean development is used the least (16%) by the participants working in the companies serving the military and defense sector; (4) Usage of waterfall is low among participants employed by small-sized companies whereas Agile/lean development is relatively popular among this class of participants; and (5) As company size increases, usage of spiral development slightly increases, whereas usage of extreme programming practices decreases.

Conclusion: The results of this exploratory study will be useful to software engineering professionals and researchers both in Turkey and world-wide by revealing the cross-factor relationship of software engineering practices versus practitioner demographics. The study raises several new research directions, e.g., (1) Why are not many practitioners using any size nor project estimation metrics and how these fundamental engineering approaches could be utilized more frequently?, (2) What are the best practices, success stories and

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challenging experiences in using SE tools?, and (3) Why is Agile/lean development is not popular in the military and defense sector and how these approaches could be utilized more frequently.

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1. Introduction

There is a wide spectrum of software engineering (SE) practices in terms of their types and maturity conducted in different software teams and companies. To understand the SE challenges in the industry and to enable researchers work on such challenges, it is important to characterize the types of SE practices and techniques used in the industry.

To characterize the type of SE practices conducted in software industry, a variety of surveys have been conducted in different countries and regions, (e.g., Beck and Perkins, 1983, Zekowitz et al., 1984, Curtis et al., 1988, Cusumano and Kemerer, 1990, Blackburn et al., 1996, Singer et al., 1997, Holt, 1997, Dutta et al., 1999, Groves et al., 2000, Cusumano et al., 2003, Aytac et al., 2003, Denger et al., 2007, Aykol, 2009, Sökmen, 2010, Egorova et al., 2010, Kirk and Tempero, 2012, Vonken et al., 2012, Garousi et al., 2014, Garousi and Zhi, 2013). For example, a recent paper (Garousi and Zhi, 2013) gathered the opinions of 246 Canadian software engineers in 2013 about software testing practices and reported various findings, e.g., (1) The importance of testing-related training is increasing over years, (2) Functional and unit testing are two common test types that receive the most attention and effort.

While the data and findings reported by existing surveys, (e.g., Beck and Perkins, 1983, Zekowitz et al., 1984, Curtis et al., 1988, Cusumano and Kemerer, 1990, Blackburn et al., 1996, Singer et al., 1997, Holt, 1997, Dutta et al., 1999, Groves et al., 2000, Cusumano et al., 2003, Aytac et al., 2003, Denger et al., 2007, Aykol, 2009, Sökmen, 2010, Egorova et al., 2010, Kirk and Tempero, 2012, Vonken et al., 2012, Garousi et al., 2014, Garousi and Zhi, 2013), are interesting, most of those studies have studied SE practices *individually*, e.g., software testing practices in Garousi and Zhi (2013). In addition to such trends and findings, it is also important to understand the *cross-factor* relationship of SE practices, e.g., what are the correlations among the degree of challenges experienced in various SE phases/tasks? E.g., if a practitioner has found requirements engineering highly challenging, it may also face challenges in software maintenance.

Furthermore, understanding and characterizing the cross-factor relationship of SE practices versus practitioner demographics are also important by asking questions such as (1) Is it the case that practitioners developing software for sectors such as military would utilize software size measurement approaches more, or use different software development methodologies, compared to practitioners developing software for other sectors?, and (2) What kinds of practices are conducted by practitioners in small versus large companies (also targeted in previous studies (Paternoster et al., 2014))? Answering these types of questions will enable researchers and practitioners to better understand and characterize SE practices and challenges in different industrial settings (e.g., large versus small companies) and then develop tailored SE solutions for each different industrial setting (e.g., various company sizes or target industry sectors).

The IT sector worldwide faces enormous challenges in delivering quality products on time and within budget (The Standish Group, 2015, The Standish Group, 2009). Ongoing challenges of the software industry in delivering projects on time and within budget led us to question the SE methods and practices used in the industry. Indeed, following proper and systematic SE practices by all the software companies across the globe is a major keystone in determining the success or failure of software projects. Understanding and characterizing the cross-factor relationship of SE practices versus practitioner

demographics will help both researchers and practitioners in improving the challenges and issues faced in the practice (Beck and Perkins, 1983, Zekowitz et al., 1984, Cusumano and Kemerer, 1990, Blackburn et al., 1996, Holt, 1997, Dutta et al., 1999, Groves et al., 2000, Cusumano et al., 2003, Aytac et al., 2003, Denger et al., 2007, Aykol, 2009, Sökmen, 2010, Kirk and Tempero, 2012, Garousi and Zhi, 2013, Garousi et al., 2013, Garousi et al., 2015, Geras et al., 2004, Garousi and Varma, 2010).

The data for this study has been collected in Turkey, a nation which is a member of the G-20 and has a vibrant software industry (Güneş, 2010). The survey performed in Turkey could provide insight for countries where software industry has grown rapidly and domestic software companies have taken a major role in developing and providing products and services to not only the domestic ICT market, but have also exported them other countries world-wide. As of 2011, there were about 1600 software development companies in Turkey (Tirpançeker, 2011). The revenue generated by the Turkish software market reached 1.8 Billion USD as of 2011 and is expected to grow several fold by 2020 (Güneş, 2010). Similar to the worldwide IT sector, while being successful, the Turkish software sector also faces certain number of challenges in developing high-quality software systems under time and cost constraints (e.g., Garousi et al., 2014, Garousi et al., 2013, Guceglioglu and Demirs, 2005). Although this study has been conducted in the context of Turkey, it can easily be replicated in other contexts and countries.

The goal of the survey reported in this paper is to understand and characterize the cross-factor relationship of SE practices versus practitioner demographics using an exploratory approach (Runeson and Höst, 2009). The goal is to benefit both SE professionals and also researchers, for observing the latest trends in the SE industry and raising new research directions specific to practitioner demographics, and also encouraging more academia-industry collaborations.

Building on top of our track-record on conducting and reporting similar surveys in SE in Canada (Garousi and Zhi, 2013, Garousi and Varma, 2010) and Turkey (Garousi et al., 2013, Garousi et al., 2015), and also our international industrial connections, e.g., (Garousi, 2010, Garousi, 2011, Garousi, 2013, Garousi, 2014), we planned, designed, and conducted a survey in year 2013 across Turkey which received responses from 202 participants. We reported in a recent study (Garousi et al., 2013) the *individual* analysis of the SE practices based on the survey data, e.g., in terms of types of software developed, the top categories are: business applications software, safety-critical and mission-critical software, and web applications. With a different goal, we analyze the cross-factor relationship of SE practices in this paper.

The remainder of this paper is structured as follows. Background and a survey of the related work are presented in Section 2. We present an overview of our Turkish-wide survey and describe the research method (design of the exploratory study) in Section 3. In Section 4, we present and analyze the survey's results. Section 5 summarizes the findings and discusses the lessons learned. Finally, in Section 6, we draw conclusions, and suggest areas for further research.

2. Background and related work

We discuss in this section the followings:

- Previous studies which have conducted cross-factor analysis of SE practices
- A brief review on the state of the software industry in Turkey

Table 1

A summary of a few studies which have reported cross-factor analysis of SE-related aspects and practices (sorted by publication year).

References	Year	Cross-factor analysis of ...	Region/country
(Blackburn et al., 1996)	1996	Correlation of project-related factors (such as ratio of time spent on each Software Development Life-Cycle phase, or team size) and the % of reduction in product development time	World-wide
(Herbsleb et al., 2001)	2001	Modification-request delay and the following factors: # of people involved, magnitude of change (diffusion), size of change, and severity	UK, German, India
(Jiang et al., 2004)	2004	Software development process maturity (i.e., based on CMMI) and 15 project performance factors (e.g., effective control over project costs, effective control over project schedules, and adherence to audit and control standards)	World-wide
(Monden, 2006)	2006	Project outsourcing ratio, function point sizes, duration, team size and productivity	Japan
(Chow and Cao, 2008)	2008	Five types of factors (organizational, people, process, technical and project) and their impact on perceived success of Agile software development projects	25 countries
(Dagenais et al., 2010)	2010	Features that characterize a project landscape, together with key orientation aids and obstacles newcomers	A group of IBM employees in North America
(Harter et al., 2012)	2012	Whether software process improvement would reduce the severity of defects	USA
This study	2014	12 SE practices and demographics-related aspects, e.g., is there any cross-factor correlation between company size and industry sector (e.g., banking, military and defense)? i.e., are larger companies focusing on specific industries?	Turkey

2.1. Cross-factor analysis of software-engineering practices

Worldwide, there exist a number of studies which have conducted cross-factor analysis of SE practices and other aspects (e.g., demographics of practitioners, companies or project). Most of those studies are based on data gathered via questionnaire-based surveys. Studies cited in Blackburn et al. (1996), Herbsleb et al. (2001), Jiang et al. (2004), Monden (2006), Chow and Cao (2008), Dagenais et al. (2010), Harter et al. (2012) are selected examples. A summary of these works is shown in Table 1 and we briefly review them next.

Based on a global survey of software developers, Blackburn et al. (1996) analyzed the factors impacting the speed and productivity of software development and suggested recommendations for improving those aspects. An example cross-factor analysis in that study was the correlation of project-related factors (such as ratio of time spent on each Software Development Life-Cycle (SDLC) phase, or team size) and the percentage of reduction in product development time.

Herbsleb et al. (2001) reported an empirical study of project delays in global software development. The authors used both survey data and data from a source code change management system to model the extent of delay in a multi-site software development organization, and explored several possible mechanisms for this delay using cross-factor analysis. To measure delays, the study used the basic tracking unit used in the projects, called the modification-request which is a request to incorporate a specific functionality into the software. To understand potential mechanisms by which multi-site work might introduce delays, they used statistical modeling techniques to build a model of the modification-request delay interval. Cross-factor analysis was applied to study the correlation and regression between modification-request delay and the following factors: # of people involved, magnitude of change (diffusion), size of change, and severity.

An exploration study of the relationship between software development process maturity (i.e., based on CMMI) and 15 project performance factors were reported in Jiang et al. (2004). The project performance factors were divided into five categories: learning, control, interaction quality, operational efficiency and flexibility. For example, the performance factors under the “control” category included: effective control over project costs, effective control over project schedules, and adherence to audit and control standards.

The report presented in Monden (2006) analyzed and mined collaboration patterns in industrial software projects. A comprehensive cross-factor analysis of various factors (including project outsource-

ing ratio, function point sizes, duration, team size, productivity) was reported.

The study reported in Chow and Cao (2008) conducted a survey study of critical success factors in Agile software projects. The study conducted cross-factor and multiple regression analysis on five types of factors (organizational, people, process, technical and project) and their impact on perceived success of Agile software development projects.

In Dagenais et al. (2010), the authors performed a grounded-theory-based study with 18 newcomers to a set of projects with the following context. When developers join a software development project, they should become familiar with various project properties and aspects (the so called “landscape features”). In such a context, the authors identified the main features that characterize a project landscape, together with key orientation aids and obstacles. The authors then conducted cross-factor analysis of those factors, e.g., to what extent poor assistance/mentoring culture would contribute to long IDE installation by newcomers?

Last but not the least, the study reported in Harter et al. (2012) investigates whether software process improvement would reduce the severity of defects. This field study analyzed a detailed data set consisting of 7545 software defects that were collected in a software project developed over a 20 year period by the systems integration division of a large information technology corporation. The software had 7.7 million lines of code developed as part of an inventory management information system. The study revealed that higher levels of software process improvement significantly reduce the likelihood of high severity defects.

The classification of the cross-factor analysis conducted in this study has also been shown in Table 1. We can see that, in summary, the aspects we cross analyze in this work are novel and will help researchers and practitioners to gain cross characterization of the selected SE practices and company demographics, as discussed in Section 1.

2.2. State of the software industry in Turkey

To provide a review on the state of the Turkish software industry, we provide a brief summary of the existing reports and articles surveying this topic, (i.e., Güneş, 2010, Tirpançeker, 2011, Turkish Software Industry Association (YASAD), 2009, Akkaya et al., 2012). We should note that we have separated sources which have explicitly conducted surveys on SE practices in Turkey from the articles only

Table 2

A summary of surveys of the SE practices in Turkey (sorted by year of study).

References	Year	Topics covered	Authors/Organization	Number of respondents
(Aytaç et al., 2003)	2001	A SPICE-Oriented, SWEBOK-Based Software Process Assessment on a National Scale: Turkish Software Sector Survey Software requirements Project management, configuration management, subcontract management and process management verification and validation, risk management, quality assurance	Specialist Group for Software, Turkish Society for Quality	68
(AKMAN and YILMAZ, 2008)	2008	Innovative capability, innovation strategy and market orientation an empirical analysis in Turkish software industry	G. Akman and C. Yilmaz	178 companies
(Aykol, 2009)	2009	SE and software management practices in Turkey Similar to the 2003 survey (Aytaç et al., 2003)	Meriç Aykol, Bahçeşehir University, Turkey	75
(Sökmen, 2010)	2010	Competency level of the software industry in Turkey Guidelines for enhancement of companies and the sector	Nermin Sökmen, The Scientific and Technological Research Council of Turkey (acronym in Turkish: TÜBİTAK)	450 sample organizations
(Turkish Testing Board, 2013, Turkish Testing Board, 2012, Turkish Testing Board, 2011, Turkish Testing Board, 2014)	2011–2013	Series of the Turkey software quality reports	Turkish Testing Board	Not disclosed
(Koch et al., 2014)	2014	From year 2011 until 2014 so far Influence of context and culture on managerial perceptions and practices in the Turkish software industry	S. Koch, A. Bener, T. Aytac, A. T. Misirli, and E. Bernroider	28 companies

expressing the state of the software industry in Turkey, without getting into technical details of SE practices (e.g., how companies conduct requirements engineering). The former group of papers is discussed in this section, while the latter group is discussed in Section 2.2

The report provided in [Turkish Software Industry Association \(YASAD\) \(2009\)](#) was put together by the Turkish Software Industrialists Association (acronym in Turkish: YASAD, www.yasad.org.tr) and discusses the software sector as the new strength of the economy. A few highlights of this report are as follows. There is a major demand for software products developed inside the country due to national security and other specific requirements (e.g., language). According to another report ([Turkey Informatics Industry Association \(TUBISAD\), 2015](#)) by the Turkey Informatics Industry Association (acronym in Turkish: TÜBİSAD), the Turkish software market has around 8% and 27% shares in the national ICT and IT markets, respectively. These shares are similar to the averages of European countries.

Another report ([Güneş, 2010](#)), put together again by YASAD, provided an overview of the software industry in Turkey. The report highlighted that the Turkish software industry has the best availability scores for qualified engineers and IT skills among Eastern European countries. Software export is featured as a significant factor in software market volume where Germany, USA and UAE are reportedly the leading export countries for Turkey. The author claims that there exists further export opportunities for Turkey via its cultural and geographical proximity to European Union, North Africa, Middle East and Turkic nations in the Central Asia.

According to [Tirpançeker, \(2011\)](#), as of year 2010, the Turkish software industry was worth about 690 million USD. According to 2010 figures, Turkish software companies exported their software products to more than 50 countries in the volume of 250 million USD. These figures are due for changes in age demographics with a 51% of the population being under 25 and more than 1.5 million Turkish Small-Medium Enterprises (SMEs) consuming IT systems in Turkey.

According to a report prepared by [Akkaya et al. \(2012\)](#) in the Turkish Institute of Strategic Thinking, a non-governmental organization,

there were, as of 2012, around 1600 software development companies in Turkey with Turkey as their headquarters. There are also many foreign (non-Turkish) software development companies who have offices in Turkey, such as IBM. On a broader scope, there are 10,000+ companies/organizations with IT-based operations in Turkey.

There are three major national associations related to Information Technology (IT) and SE in Turkey: (1) Turkish Software Industrialists Association (acronym in Turkish: YASAD), (2) Informatics Association of Turkey (acronym in Turkish: TBD, www.tbd.org.tr), and (3) Turkish Informatics Foundation (acronym in Turkish: TBV, www.tbv.org.tr) which monitor the state of the industry and organize events in this area. There are also more focused SE-related associations such as the Turkish Testing Board (a.k.a., Turkish software testing and quality association, www.turkishtestingboard.org).

There are annual national conferences and symposiums related to IT and SE, the main ones being: the Turkish National Software Engineering Symposium (acronym in Turkish: UYMS, www.uyms.org.tr), and the National Informatics Symposium (www.citex.org).

Our literature search also identified a number of surveys on SE practices in Turkey (Aytaç et al., 2003, Aykol, 2009, Sökmen, 2010, Turkish Testing Board, 2013, Turkish Testing Board, 2012, Turkish Testing Board, 2011, Turkish Testing Board, 2014, Koch et al., 2014, AKMAN and YILMAZ, 2008) which have been briefly summarized in Table 2 (sorted by year of study).

3. Research method

We conducted recently (in 2013) a national survey of SE practices in Turkey. The survey goal was to characterize the SE practices in Turkey for the purpose of identifying the trends, and also to provide a view on the latest SE techniques, tools and metrics used by practitioners (professionals) and the challenges faced by them to benefit both SE professionals and also researchers both in Turkey and world-wide. The survey had 46 questions (reviewed in Section 3.1).

In the papers which have resulted from our Turkey-wide survey (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we have

analyzed data for each of the survey questions individually. However we were not able to answer questions on the relations between SE practices and practitioner demographics including their companies and projects. For example, two of the 46 survey questions were the following: target industry sector (e.g., banking, military and defense) and usage of size measurement approaches. While we reported each of the above aspects in Garousi et al. (2014), Garousi et al. (2013), Garousi et al. (2015), it is important to also assess the correlation between the target industry sector and use of size measurement approaches, and see whether practitioners developing software for certain sectors such as military would utilize different software size measurement approaches compared to practitioners developing software for other sectors.

To enable the reader to better grasp the contributions of this paper, we briefly review our recent Turkish-wide survey in Section 3.1. Afterwards, we present in Section 3.2 the design of the current exploratory study (its goal and research questions).

3.1. Brief review of our recent Turkish-wide survey

We systematically designed the online survey based on our past survey experience in the Canadian context (Geras et al., 2004, Garousi and Varma, 2010). We also benefited from the SWEBOK (version 2004) (Bourque and Fairley, 2004) in categorizing our survey and its questions. The 2004 version of the SWEBOK divides the SE discipline into 10 knowledge areas which correspond, in a one to one relationship, to the research questions of the survey.

The criteria we used in designing the survey's question set was to make sure that the questions were as relevant to the industry as possible and also to capture the most useful information. In order to develop a survey that would adequately cover the latest topics in SE while at the same time permitting us to minimize the number of questions, we reviewed the similar past surveys (Beck and Perkins, 1983, Zerkowicz et al., 1984, Cusumano and Kemerer, 1990, Blackburn et al., 1996, Holt, 1997, Dutta et al., 1999, Groves et al., 2000, Cusumano et al., 2003, Aytac et al., 2003, Denger et al., 2007, Aykol, 2009, Sökmen, 2010, Kirk and Tempero, 2012) and designed a draft set of questions. The authors then consulted with several industrial practitioners to do a careful peer review on the draft set of questions. Getting feedback from industrial partners in design of surveys is an approach followed in previous surveys as well (e.g., Geras et al., 2004, Garousi and Varma, 2010). The goal behind this phase in our survey design was to ensure that the terminology used in our survey was familiar to the participants. This step was necessary since, unfortunately, the SE terminology used in academia versus industry can sometimes be slightly different or even confusing. The feedbacks from the industrial practitioners were used to finalize the set of survey questions.

After the iterative design and improvement of the questions, we finally had 46 questions. The complete list of the questions used in the survey is shown in the appendix of this paper (Table 16).

The survey was hosted on an online survey hosting service called SurveyMonkey (www.surveymonkey.com). Research ethics approval for the survey was obtained from the Middle East Technical University's Applied Ethics Research Center in May 2013. According to the justifications discussed in our recent works (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we used convenience sampling method, a type of non-probabilistic sampling method where subjects are selected due to their convenient accessibility to the researchers, in this study. We sent email invitations to our network of partners/contracts in Turkish software companies and asked them to forward the invitation email to their own contacts in order to maximize the outreach of the survey. We also made public invitations to the Turkish SE community by posting messages in social media, e.g., LinkedIn, Yahoo Groups and Twitter. The survey was available to participants for three months during the sum-

mer of 2013. Participants were asked to complete the survey online and participation was voluntary and anonymous. Respondents could withdraw their results from the survey at any time and, as per the ethics guidelines, researchers agreed to publish only summary and aggregate information from the survey. At the end of data collection, the survey included data from 202 software practitioners across Turkey.

In terms of descriptive statistics of the dataset gathered via the survey, as reported in Garousi et al., (2014), Garousi et al. (2013), Garousi et al. (2015), most of the 202 survey participants were "software developers" and "software engineers". The average and median values of participants' years of work experience were 7.8 and 6 years respectively. About 40% of the respondents had a degree in computer engineering and 54% and 40% of respondents had a Bachelor's and a Master's degree, respectively. In terms of geographical distribution, most participants were located either in Ankara (67%) or Istanbul (24%).

There was a good mix of participants working in various software organizations. Most common target sector of the organizations employing the participants were Military and Defense, IT and Telecommunication, and the Government (not including military and defense). More than 80 of the total of 202 participants were employed by large organizations (with 500+ employees), but still there was a good mix from other ranges. More detailed demographics of the participants and further details about the survey's design, questions and execution can be found in Garousi et al. (2014), Garousi et al. (2013), Garousi et al. (2015).

3.2. Design of the exploratory study

We discuss in this section goal, analysis approach, research questions and the survey's raw data.

3.2.1. Goal and analysis approach

The two research approaches that we use in this study are the Goal, Question, Metric (GQM) methodology (Basili, 1992) and the exploratory case study approach (Runeson and Höst, 2009). Stated using the GQM's goal template (Basili, 1992), the goal of this study is:

- To conduct cross-factor analysis of SE practices and practitioner demographics for the purpose of being able to better understand and characterize SE practices and challenges in specific industrial settings (e.g., large versus small companies or various target industry sectors) and, subsequently, to develop tailored SE solutions for each class of practitioners (e.g., employed by companies with different size or target industry sector).
- And to benefit both SE professionals and also researchers both in Turkey and world-wide, for observing the latest trends in the SE industry and identifying the areas of strength and weaknesses and encouraging more academia-industry collaborations.

The study approach is exploratory in the sense that we aim at finding out, in an exploratory manner, the cross-factor relations among SE practices and practitioner demographics including their companies and projects. Such an analysis will enable us to seek new insights and generating ideas and hypotheses for future studies. Based on the above study goal (phrased using the GQM approach), we will raise a set of 12 Research Questions (RQs) in the next section below. Each RQ will then be addressed in Section 4 using a set of proper metrics.

To investigate the potential factors that could correlate with other factors, we created a two-dimensional matrix where the individual survey questions of the recent survey (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015) are listed in both rows and columns (only demographics), as shown in Table 3. Note that, due to being somewhat similar, questions 40 and 41 of the survey, and also questions 45 and 46 were merged. As a first step in our investigation, we aimed at conducting the cross-factor correlation analysis among

Table 3Cross-factors correlations there were chosen as the most important ones to be investigated^a.

Question	Description	Correlation with itself	Demographic aspects (survey questions)										
			1	2	3	4	5	6	7	8	9	10	11
			Participant profile					Company profile		Software projects profile			
			Position	Years of work experience	Academic degree	University degree	City (location)	Target sector (gov-ernment, etc.)	Company size	Type of software (general versus specific industry)	Type of software (safety-critical, etc.)	Usage of size measurement	Number of software projects
1	Current position(s)			x	x								
2	Years of work experience												
3	Highest academic degree												
4	University degree												
5	Work city												
6	Target sector of your company								x				
7	Company size (number of employees)						x						x
8	Software type (in-house, etc.)											x	
9	Software type (safety-critical, etc.)												
10	Size measurement methodology							x					
11	Number of projects in past two years												
12	Effort spent on each SDLC phase								x				
13	Participants' involvement in phases			x						x	x		
14	Level of challenge in each SDLC phase	x		x	x				x				
15	Software development methodologies	x							x		x		
16	Following process improvement models							x	x				
17	Practices - software engineering process	x						x	x				
18	Practices - requirements engineering							x		x			
19	Notations used for software requirements					x		x					
20	Practices – software design	x						x					
21	Types of design activities							x					
22	Design-related quality attributes		x	x				x					
23	Programming languages												
24	Practices – software development	x		x				x	x				
25	Practices – software testing							x					
26	Phasing of testing during the SDLC							x	x				
27	Test types/levels	x						x	x				
28	Test-case design techniques	x			x			x					
29	Test automation			x				x	x				
30	Test-related metrics			x				x	x				
31	Other quality-related metrics							x	x				
32	Ratio of testers to developers								x				
33	Criteria for terminating testing							x					
34	Types of software maintenance	x											x
35	Level of challenges during maintenance												
36	Description of maintenance challenges												
37	Type of challenges during maintenance							x					
38	Practices - product release and delivery								x				
39	Practices - product support								x				
40/41	Practices - software project management			x				x	x				
42	Usage of software engineering tools	x		x				x	x				x
43	Software quality practices							x					
44	Dedicated R&D departments/units							x	x				
45 / 46	Interacting with researchers and reading SE papers and articles				x	x		x	x				

^a Cells marked with 'x' denote the correlation combinations that we found to be the most interesting, relevant and meaningful. The grayed cells marked with 'x' identify the combinations selected to be investigated in this study.

the SE practices (questions 12–46) with the profiles and demographic characteristics (questions 1–11). Thus, we only listed questions 1–11 in the columns, but all the questions have been listed in the rows.

Since we have 45 rows and 11 columns, we could potentially have 495 (45×11) cross-factor correlation combinations. However, we carefully reviewed each case to ensure choosing the most relevant ones, with reasonable “face validity” (Procaccino et al., 2005), and those which would be expected to reveal useful and meaningful findings. For example, the hypothesis corresponding to row #6 and column #7 (Q6-7) in Table 3 was formulated as follows: Is there any correlation between company size and industry sector? i.e., are practitioners employed by larger companies focusing on specific industries?

Out of all possible 495 cross-factor combinations, using the expert opinion of the authors and in consultation with a few industry contacts, we found 75 combinations to be the most interesting, relevant and meaningful ones. These combinations have been labeled by ‘x’ in the cells of Table 3. Note that the notion of correlation here is symmetric, i.e., correlation of factor A and factor B is the same as correlation of factor B and factor A. Thus, we do not consider the duplicate cases.

The third column of Table 3 is labeled as “correlation with itself”, denoting the auto-correlation of practices in a question with itself, e.g., for question #17 (process-related practices), we raised the following research questions (RQ): Is there any correlation between the four process-related practices (using standard processes, systematic project management, systematic monitoring, and learning from past projects) with one another? i.e., do participants use any-pair of them together?

3.2.2. Research questions

Due to space constraints, we study in this article only 12 out of all 75 RQs (marked with ‘x’ in Table 3) that were found to be the most interesting. These combinations have been identified with grayed cells in Table 3. The other remaining RQs have been planned to be investigated in future works.

Based on the study goal raised in the above section (following the GQM approach), for each of the cross-factor combinations, we formulated a corresponding RQ as listed below. These RQs will be addressed in Section 4. The RQs have been grouped under five categories.

Cross-factor analyses between SE practices and participants’ company demographics

- RQ 1. Q10-6 (correlation between questions 10 and 6 of the survey): How does usage of different software size measurement approaches (e.g., lines of code and function points) differ in various companies employing participants w.r.t. their target industry sectors (banking, government, military, etc.)? This RQ is a “descriptive-comparative” type of a RQ, as per the RQ classification proposed by Easterbrook et al. (2008). In other words, what is the cross-factor correlation between use of software size measurement approaches and the company’s target industry sector? One would expect that practitioners developing software for sectors such as military would utilize software size measurement approaches more compared to practitioners developing software for other sectors.
- RQ 2. Q15-6: How does usage of different software development methodologies differ for practitioners employed by various companies with different target industry sectors? This RQ is also a “descriptive-comparative” type (Easterbrook et al., 2008) of a RQ. In other words, what is the cross-factor correlation between use of software development methodologies and the company’s target industry sector? One would expect that practitioners working for companies developing software for sectors such as military would utilize different software development methodologies compared to

sectors such as business applications, e.g., Waterfall versus Agile.

- RQ 3. Q15-7: What is the cross-factor correlation between use of software development methodologies and practitioner employer company size (number of employees)? One would expect that larger companies would utilize different software development methodologies compared to smaller companies, e.g., Waterfall versus Agile.
- RQ 4. Q22-6: What is the cross-factor correlation between the importance of design-related quality attributes and companies’ target sector? One would expect that, for practitioners employed by companies developing software for specific sectors such as military, certain design-related quality attributes would be more important compared to others.
- RQ 5. Q27-7: What is the correlation between the test types/levels conducted and the company size? Are practitioners working for larger companies conducting more test types/levels?

Cross-factor analyses between SE practices and participants’ project demographics

- RQ 6. Q34-11: What is the correlation between ratios of software maintenance activities versus number of projects? One would expect that, if a practitioner employed by a company having more projects at the same time, s/he will conduct more software maintenance activities.
- RQ 7. Q42-11: What is the correlation between usage of SE tools versus number of projects? One would again expect that, if a company has more concurrent projects, it would more likely use more SE tools, e.g., for requirements engineering and project management.

Cross-factor analyses between SE practices and participant demographics

- RQ 8. Q14-2: Does more work experience of a practitioner imply that s/he will face fewer challenges in SDLC phases and SE tasks? In other words, what is the correlation between the level of challenges experienced by a practitioner in SDLC phases and SE tasks versus her/his years of work experience? One would again expect that, if a practitioner has more years of work experience, s/he would experience less challenges in SDLC phases and SE tasks.
- RQ 9. Q42-2: Does more work experience of a practitioner imply more usage of SE tools by the practitioner? In other words, is the extent to which a practitioner uses SE tools correlated with his/her years of work experience? It may be the case that, the more years of work experience a practitioner has, the more likely s/he will use SE tools in her/his tasks.

Cross-factor analyses among SE practices

- RQ 10. Q14-14: What are the correlations among the degree of challenges experienced in various SE phases/tasks? E.g., if a participant has found requirements engineering highly challenging, it may also face challenges in software maintenance.
- RQ 11. Q15-15: To what extent different software development methodologies are used (or not used) by the same participant? For example, does wide adoption of the waterfall methodology means narrow adoption of Agile development?

Cross-factor analyses among demographics

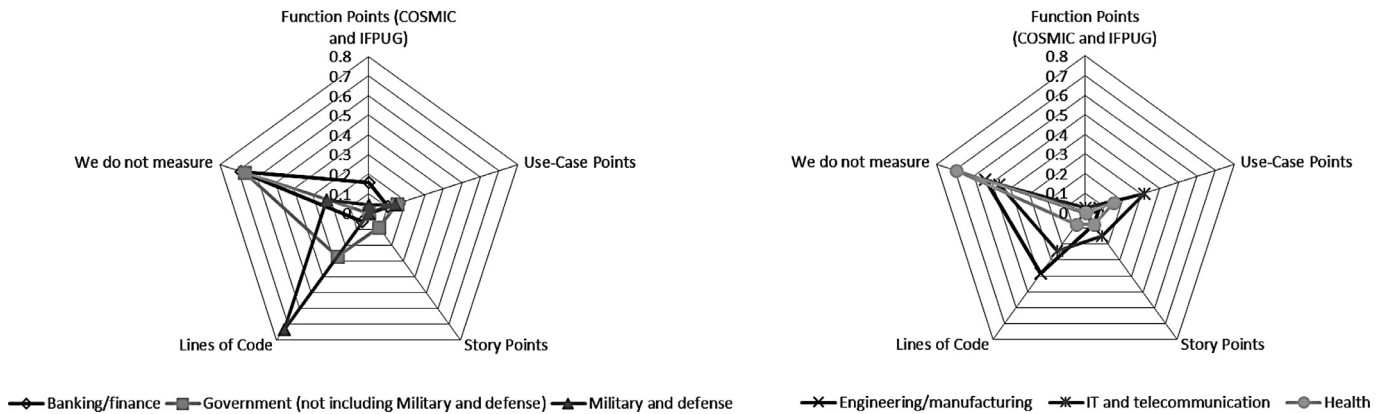
- RQ 12. Q6-7: What is the cross-factor correlation between company size and industry sector (e.g., banking, military and defense)? i.e., could we say that larger companies are focusing on specific industries?

In designing and framing the RQs, to ensure the rigor and usefulness of our results, we have used the RQ classification and guidelines

Table 4

A classification scheme for RQs as proposed by Easterbrook et al. (2008).

RQ category	Sub-category	RQ code	Example
Exploratory	Existence	E	Does X exist?
	Description and classification	DCL	What is X like?
	Descriptive-comparative	DCO	How does X differ from Y?
Base-rate	Frequency distribution	FD	How often does X occur?
	Descriptive-process	DP	How does X normally work?
Relationship	Relationship	R	Are X and Y related?
Causality	Causality	C	Does X cause (or prevent) Y?
	Causality-comparative	CC	Does X cause more Y than does Z?
	Causality-comparative interaction	CCI	Does X or Z cause more Y under one condition but not others?
Design	Design	D	What's an effective way to achieve X?

**Fig. 1.** Use of size measurement approaches versus target industry types.

proposed by Easterbrook et al. (2008) as shown in Table 4. For example, RQs 1 and 2 are “descriptive-comparative”. The other RQs are usually of type “relationship” since they aim at exploring cross-factor correlations between a given pair of aspects.

3.2.3. Survey's raw data and ensuring replicability

Replication and replicability of empirical studies are very important for any empirical science such as SE (Louridas and Gousios, 2012, Shull et al., 2007). However, according to Louridas and Gousios (2012), these important aspects are regrettably have not been a priority for many SE researchers and many published SE studies have not taken them into account.

To ensure replicability of our results and also to encourage further studies in this area, we provide all the raw data gathered during our survey studies (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015) and also processed during this work in an Excel file available publicly online (Garousi, 2014).

4. Results

The results of analysis for each selected research question in the study are reported in this section.

4.1. Cross-factor analyses between SE practices and participants' company demographics

RQs 1–5 are discussed in this section.

4.1.1. RQ 1: use of size measurement approaches versus target sector type (Q10-6)

RQ 1 asked about the cross-factor correlation between use of software size measurement approaches and the company's target industry sector. One would expect that practitioners employed by companies developing software for sectors such as military would utilize software size measurement approaches more compared to

practitioner developing software for other sectors. Some of the underlying reasons for such differences could be: (1) based on target industry sector of a company, if the company is CMMI certified, it would be required to use formal size measurement approaches, and (2) there has been evidence in other regions, e.g., USA, that companies in the defense industry tend to utilize software size measurement approaches more, e.g., according to two special issues (CrossTalk: The Journal of Defense Software Engineering, 2014, CrossTalk: The Journal of Defense Software Engineering, 2014) of the CrossTalk (the Journal of Defense Software Engineering) focused on software measurement.

Fig. 1 shows usage rates of software size measures by participants employed by organizations from different target sectors as a radar chart. In the analysis, we only considered six most popular target sectors that were represented with at least 20 data points in our dataset. Excluded target sectors included: business, management, insurance sectors and “other” category that included food, e-commerce, automotive, textile, game technologies, education, transportation and logistics. For brevity and readability, data have been placed in two radar charts. In our survey questionnaire (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we had provided a list of five pre-listed size measurement approaches for respondents to choose from: lines of code (LOC), COSMIC function points, IFPUG function points, use-case points, and story points. If respondents wanted to report additional metrics not present in the above list, they could do so in the “other” field which was available in the questionnaire.

We can observe in Fig. 1 that more than 55% of participants employed by companies operating in each sector do not measure software size at all, except for the participant organizations developing military and defense-related software for which this ratio is only 22%. The LOC metric is a popular size measure in military and defense sector with a usage rate of 73%. The LOC metric is also used to some extent by participants working in companies producing for engineering, manufacturing, government, IT and telecommunication sectors with usage rates of 38%, 27% and 24% respectively.

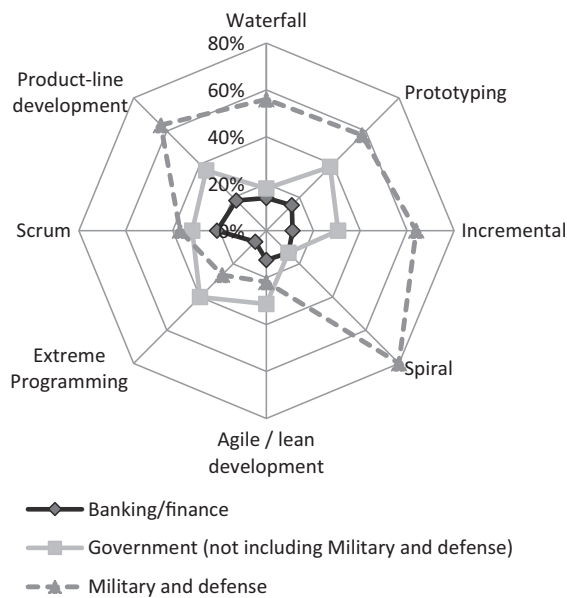


Fig. 2. Usage of software development methodologies versus target industry sector.

Point-based estimation metrics (i.e., COSMIC FP, IFPUG FP, use-case points and story points) are used rarely. Among these measures, use-case points are used slightly more, especially in IT/telecommunication sector (31%). Functional size measures (COSMIC and IFPUG) are almost not used in any of the target sectors. Thus, we can imply that these metrics are not popular in the Turkish SE industry.

Our findings on many practitioners not using any size measure and not using measures for project estimation might be factors for poor project planning and control practices which have to be investigated in future studies. Also, future studies are needed to study the root causes of why formal size measurement is not usually conducted by practitioners.

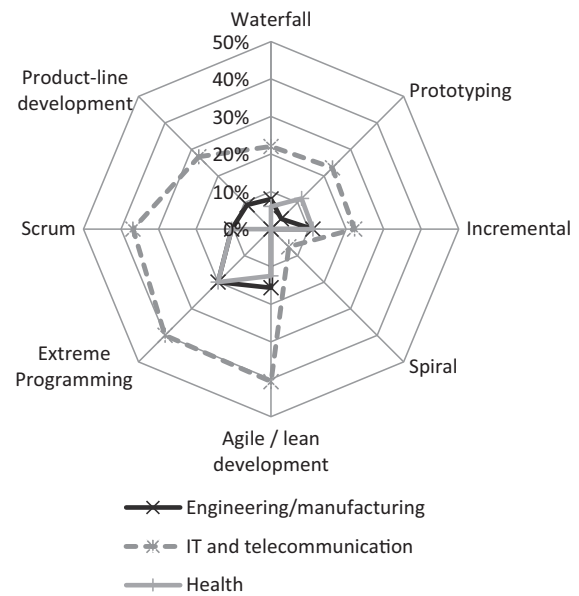
4.1.2. RQ 2: usage of software development methodologies versus target industry sector (Q15-6)

RQ 2 asked about the cross-factor correlation between use of software size measurement approaches and target industry sector of the companies that the respondents work in. One would expect that companies developing software for sectors such as military may utilize different software development methodologies (e.g., waterfall versus Agile) compared to companies developing software for other sectors.

Fig. 2 depicts, as a radar chart, the usage of various software development methodologies by participants employed by companies performing in different industry sectors. In our survey questionnaire (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we had provided a list of eight pre-listed software development methodologies for respondents to choose from: waterfall, prototyping, incremental, spiral, Agile or lean development, extreme programming, Scrum and product-line development. If respondents wanted to report additional methodologies not in the above list, they could do so in the “other” field which was available in the questionnaire.

According to the figure; Scrum, spiral and product-line development are not very popular in any of the target sectors. Prototyping and extreme programming are also rather unpopular. Waterfall, incremental and Agile/lean development are the most popular methodologies whose usage rates do not fall below 30% in any of the sectors, except military and defense sector where Agile/lean development is used around 16% of the time.

Moreover, if we look at the sector-based distributions, we observe that, more than half of the practitioners working in compa-



nies that produce products for banking and finance sector use waterfall life-cycle. Companies targeting military and defense sector, which is the most represented sector in our group of respondents, usually follow waterfall and/or incremental development life-cycles. Agile and lean development is the most popular methodology for participants employed by companies developing software for engineering/manufacturing, IT and telecommunication and health sectors. Waterfall is again also popular in these sectors with more than 40% usage rate.

4.1.3. RQ 3: usage of software development methodologies versus company size (Q15-7)

Differences in usage of various software development methodologies with respect to company size are depicted in Fig. 3. Regression fits have been provided based on the individual-value plots of yes/no votes of the respondents, an example of which has been shown for the case of waterfall model.

We can observe that usage of waterfall is low among practitioners employed by small-sized companies whereas Agile/lean development is relatively popular among this class of practitioners. We can see noticeable trends between company size and development methodologies for spiral and extreme programming. As company size increases, usage of spiral life-cycle slightly increases, whereas usage of extreme programming practices decreases. For the other development methodologies, the trends are rather mixed, denoting that there is no clear trend in practitioners employed by various companies with different sizes. As these observations are synthesized, we infer that in larger companies waterfall-like development is still favorable whereas in smaller companies Agile/lean is preferred more.

4.1.4. RQ 4: importance of design-related quality attributes versus target industry sector (Q22-6)

Our hypothesis was whether practitioners employed by companies focusing on specific industries, e.g., defense, put a heavier emphasis on specific design-related quality attributes, compared to other practitioners. In our survey questionnaire (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we had provided a list of nine pre-listed design-related quality attributes for respondents to choose from: extensibility, fault-tolerance, maintainability, modularity, reliability, reusability, robustness, security and usability.

Fig. 4 visualizes the trends, where the Y-axis shows the “median” data point for the 5-point Likert scale denoting importance

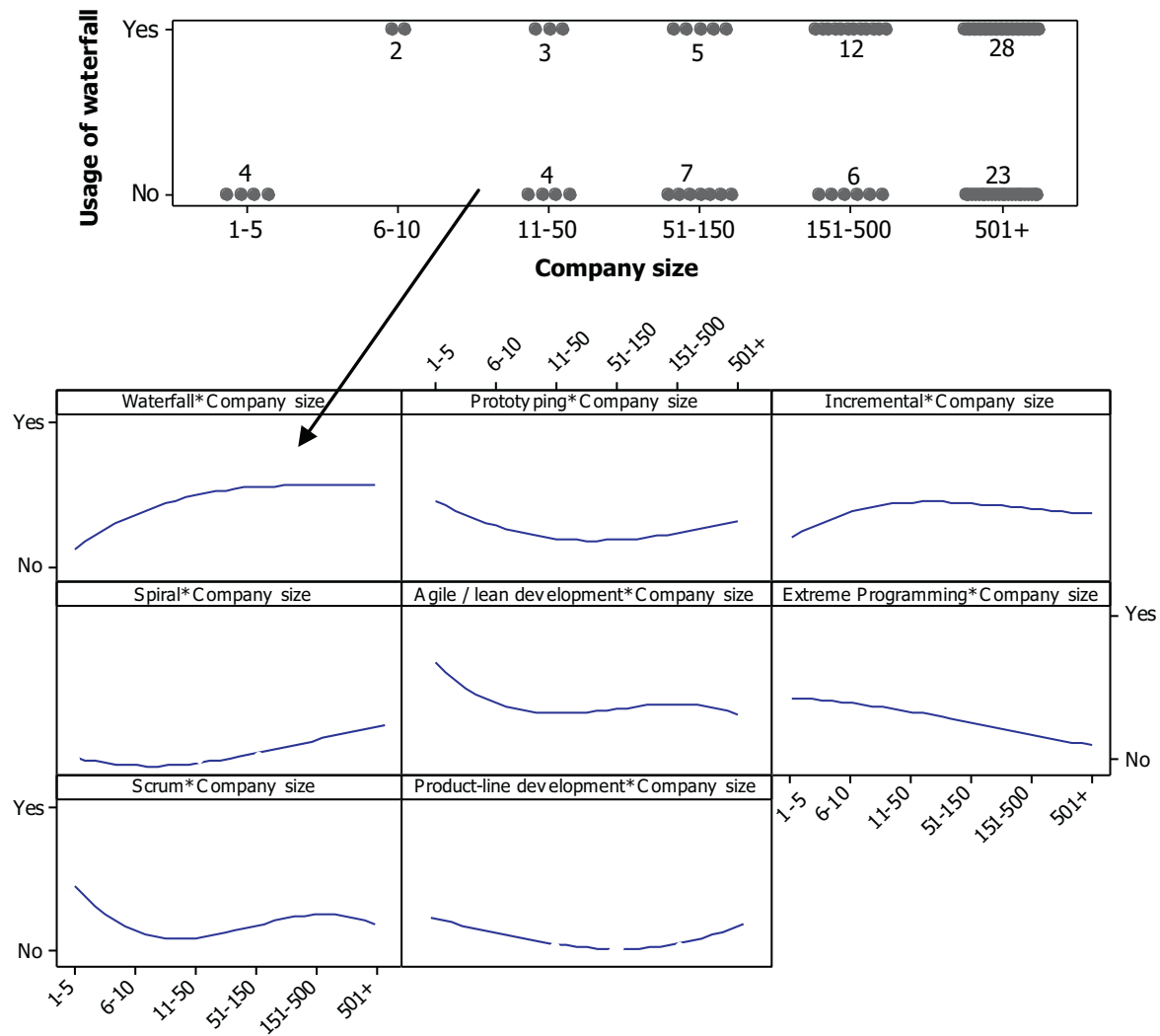


Fig. 3. Usage of software development methodologies versus company size.

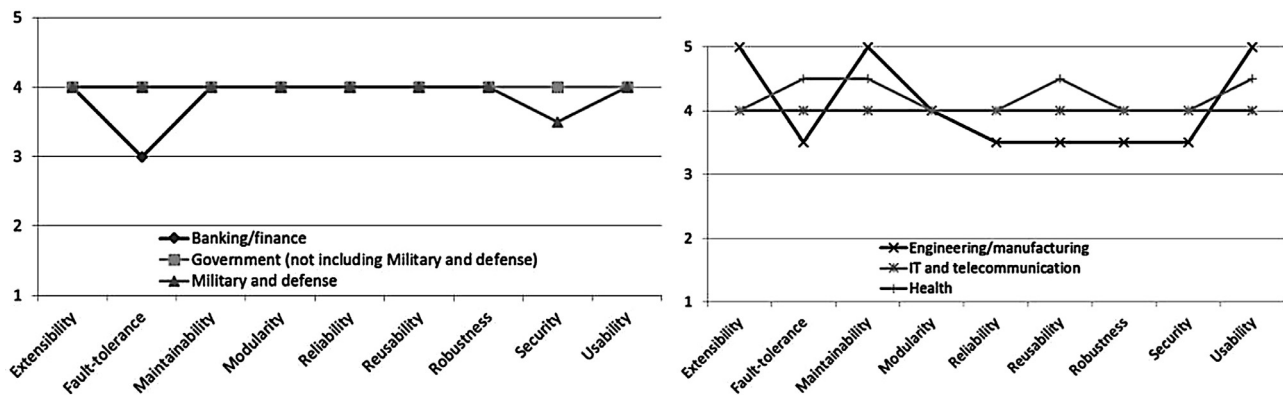


Fig. 4. Design-related quality attributes versus target sector (the Y-axis shows the “median” data points).

(emphasis) of each design-related quality attribute: never (1), seldom (2), sometimes (3), frequently (4), and always (5). We summarize our observations in the following:

- As we can observe, most of the curves are almost overlapping and generally in the range of 3–5 (corresponding to “sometimes” to “always”), denoting that for participants, regardless of their companies’ target industry sectors, different design-related qual-

ity attributes are “sometimes” to “always” important. The average values are mostly near 4, corresponding to being “frequently” important.

- The case of the companies targeting engineering/manufacturing industry sectors is quite different (i.e., lower) than others, for three of the design-related quality attributes (reusability, robustness and security). It is not easy to explain why this is the case as those quality attributes are important considerations

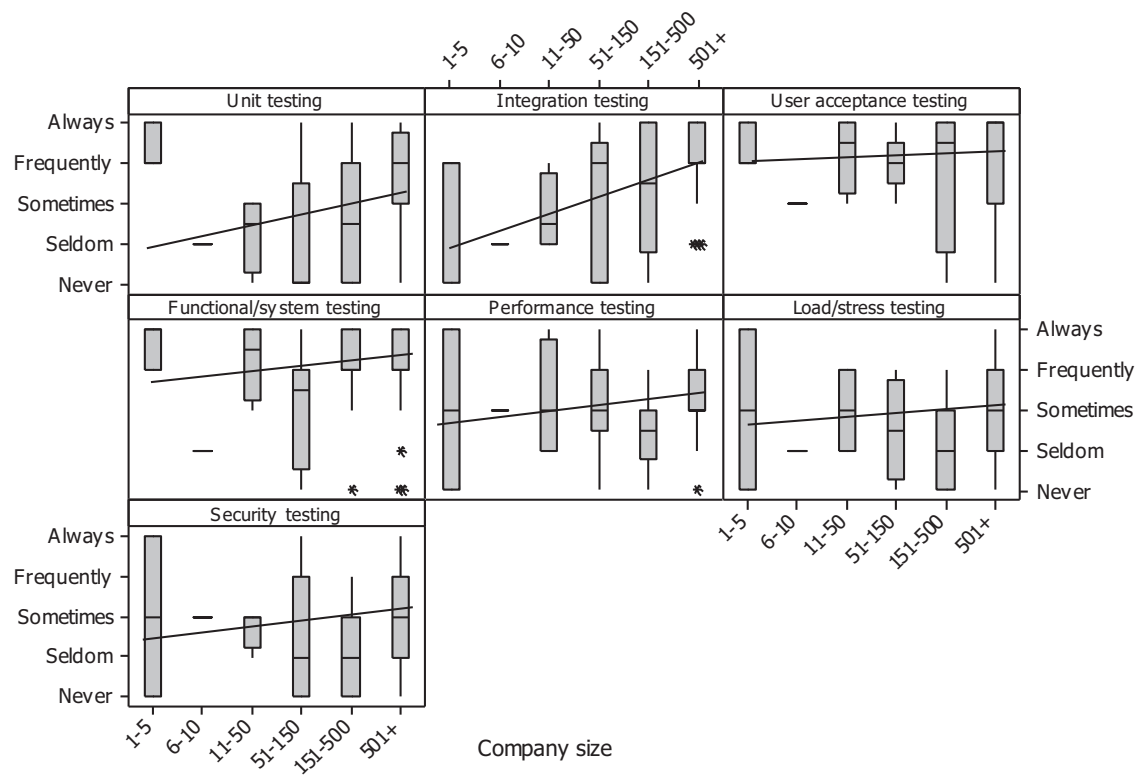


Fig. 5. Execution frequency of different test types/levels (unit testing, integration testing, user acceptance testing, functional/system testing, performance testing, load/stress testing, and security testing) versus company size.

Table 5

Spearman's rho correlation values of execution frequency of different test types/levels versus company size.

	Company size	Unit testing	Integration testing	User acceptance testing	Functional/system testing	Performance testing	Load/stress testing
Unit testing	0.27						
Integration testing	0.40	0.30					
User acceptance testing	0.10	0.07	0.51				
Functional/system testing	0.11	0.11	0.51	0.45			
Performance testing	0.16	0.44	0.35	0.11	0.20		
Load/stress testing	0.17	0.34	0.36	0.31	0.33	0.79	
Security testing	0.13	0.30	0.33	0.25	0.23	0.65	0.73

in all domains, including engineering/manufacturing software systems.

- For participants whose companies targeting engineering/manufacturing industry sectors, usability considerations seem to be very important.

4.1.5. RQ 5: test types/levels versus company size (Q27-7)

The objective of RQ 5 was to assess the correlation between the types of test types/levels conducted by a practitioner and the company size. One may expect that larger companies employing the participants may have more resources to be spent on various test types/levels (e.g., unit or load testing).

Relations between execution frequencies of different types/levels of testing versus company size are represented as box-plot charts in Fig. 5. Linear regression fit lines have also been shown. We only notice slight correlations for the case of unit, load and integration

testing. The Spearman's rho correlation values are also provided in Table 5. Note that we have used the Spearman's rho correlation metric because we are looking for correlations between interval (company size) and ordinal data (test types/levels). Also a heat-map based color coding has been used to better present the trends. All of the correlation values, except for integration testing, are quite close to zero, denoting absence of correlations among company size and different types/levels of testing.

It is interesting to observe that as company size grows, there is more chance for integration testing to be conducted perhaps denoting the higher maturity and awareness of testing practices in those companies. This could also be due to the general observation that large companies usually undertake more complex and large projects compared to smaller companies. Thus, as our finding suggests, product integration and integration testing is expected to be more critical for larger companies developing larger software products. Also,

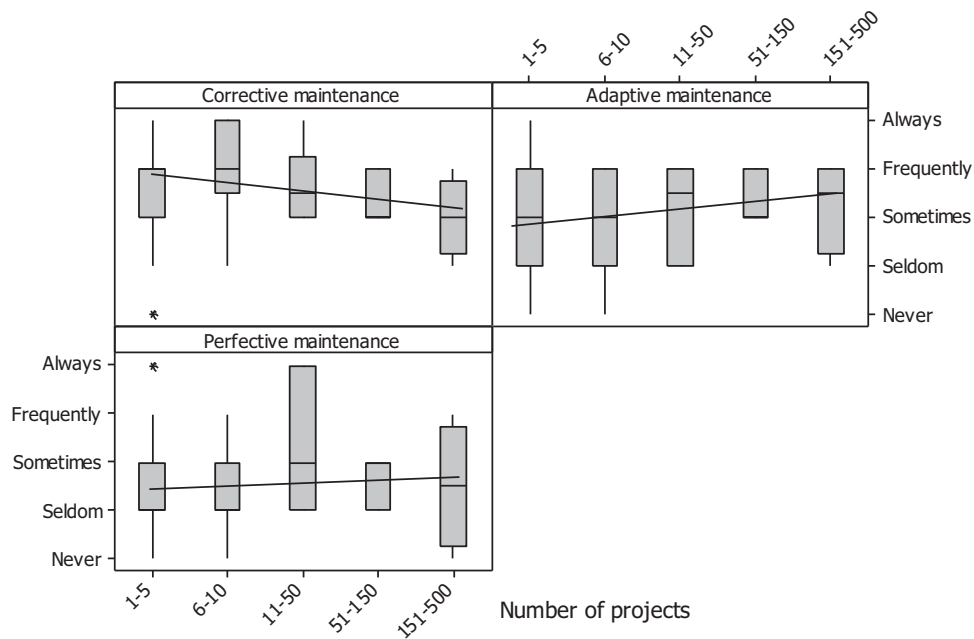


Fig. 6. Types of maintenance versus number of projects undertaken by companies.

according to Table 5, there are noticeable correlations (values > 0.5) among the different types of non-functional testing (i.e., load, stress, security and performance testing), i.e., if a company decides to conduct one of these types of testing, it would also likely conduct the other non-functional testing types.

4.2. Cross-factor analyses between SE practices and participants' project demographics

RQs 6 and 7 are discussed in this section.

4.2.1. RQ 6: types of software maintenance activities versus number of projects (Q34–11)

RQ 6 explored execution frequency of different types of software maintenance activities versus number of projects undertaken in each company employing the participants in the past two years (2011–2013). The execution frequency of the maintenance activities were chosen from a 5-point Likert scale: never (1), seldom (2), sometimes (3), frequently (4), and always (5). As it has been well established in the software maintenance literature (Grubb and Takang, 2003), three types of software maintenance activities were provided to the survey participants to solicit their opinions for:

- Corrective maintenance: reactive modification of a software product performed after delivery to correct discovered problems
- Adaptive maintenance: modification of a software product performed after delivery to keep a software product usable in a changed or changing environment
- Perfective maintenance: modification of a software product after delivery to improve performance or maintainability

One would expect that, if a company has more projects at the same time, more software maintenance activities is conducted since, in general, there will be more chances (or occurrences) of the above type of activities.

Fig. 6 presents the cross-factor relationship among the above three types of maintenance activities and number of projects undertaken by each employer company of the participants in the two year period of 2011–2013. Linear regression fit lines have also been shown. Spearman's rho correlation values of execution frequency of different maintenance activities versus number of projects (Table 6). We notice

Table 6

Spearman's rho correlation values of execution frequency of different maintenance activities versus number of projects in each company.

	Corrective maintenance	Adaptive maintenance	Perfective maintenance
Number of projects	−0.10	0.01	0.07

that for the cases of adaptive and perfective maintenance, there are weak positive correlations with increase in number of projects. There is a slight negative correlation between frequency of corrective maintenance and number of projects undertaken. This could be perhaps explained as follows: when a company has more ongoing projects, perhaps its engineers will have less time to spend on fixing defects (i.e., corrective maintenance). Note again that this is only a hypothesis and needs evaluation in future studies.

4.2.2. RQ 7: usage of SE tools versus number of projects (Q42–11)

As the next question, RQ 7 explored the correlation between usages of SE tools versus number of projects undertaken by employer companies of the participants in the two year period of 2011–2013. One would expect that, if a company has more concurrent projects, practitioners would more likely use more SE tools, e.g., for requirements engineering, issue (defect) and project management. We did not ask about usage frequency of software development tools (e.g. compilers, and IDEs) as we assumed such tools are used by all participants who work in software engineering/development roles. Fig. 7 shows the data as box plots. Linear regression fit lines have also been shown.

All the trend lines shown in Fig. 7 are in strictly increasing slope, denoting that, the higher the number of projects, there is more likelihood that respondents would use SE tools. This is logical and as expected since, as discussed above, challenging SE tasks such as requirements engineering, issue (defect) management and project management require powerful tools to be conducted effectively and efficiently. This is specially the case when there are a large number of projects in a company concurrently (e.g., over 200). Another explanation is that companies working on larger number of concurrent projects usually possess higher maturity levels (e.g., in terms of the

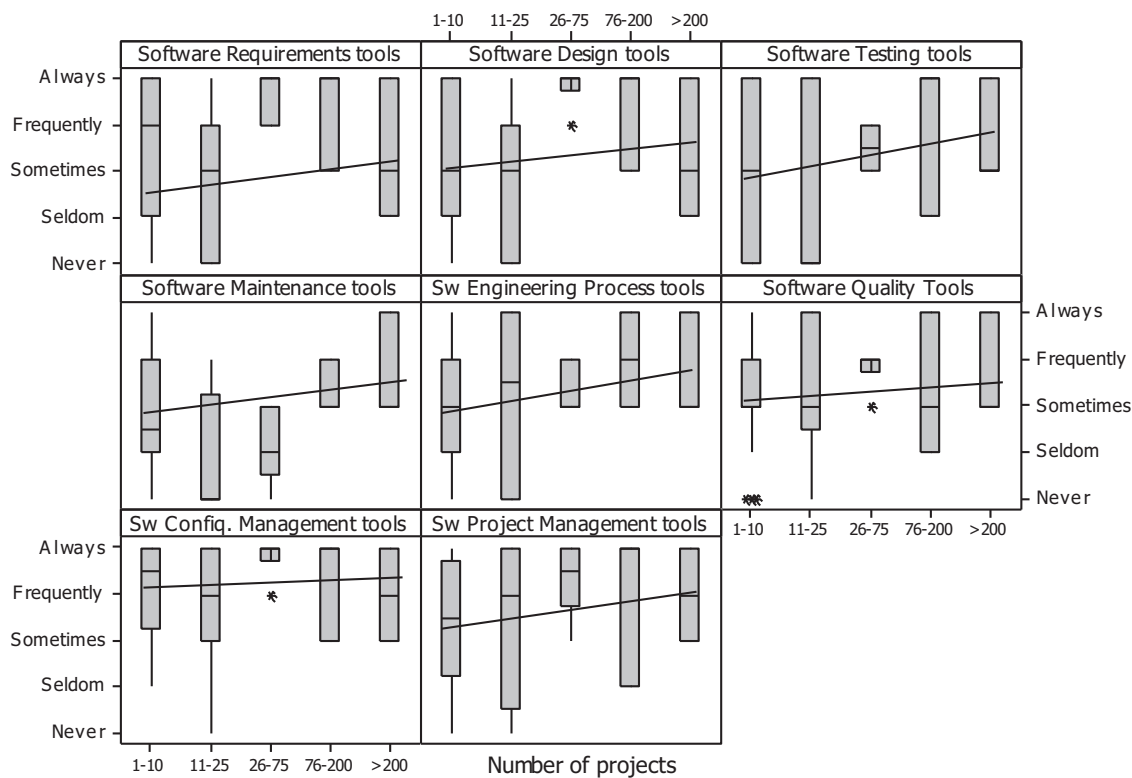


Fig. 7. Usage of SE tools versus number of projects.

CMMI) and are thus more aware of using latest industry-standard SE tools.

It is good news that the Turkish SE industry is quite mature in using SE tools. But of course, probing and exploring the success stories and challenging experiences in using SE tools in this context would be an interesting line of investigation in future studies.

4.3. Cross-factor analyses between SE practices and participant demographics

RQs 8 and 9 are discussed in this section.

4.3.1. RQ 8: challenges experienced in SE phases/tasks versus years of work experience (Q14-2)

RQ 8 aimed at exploring the correlation between the level of challenges experienced by a practitioner in different SDLC phases and SE tasks versus her/his years of work experience. One would expect that, if a practitioner has more years of work experience, s/he would experience less challenges in SDLC phases and SE tasks. In our survey questionnaire (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we had provided the list of five SDLC phases: requirements, design, implementation, test and maintenance, and a set of nine SE (support) tasks (based on the SWEBOK (Bourque and Fairley, 2004)): configuration management, project management, engineering process, tools, quality assurance, technical documentation, communication with team members, communication with management, and communication with end users.

Based on the data we received in the survey, the best way to visualize these data was an individual-value plot as shown in Fig. 8 (which includes linear regression fit lines). Spearman's rank correlation coefficient and p -value values for the same data are provided in Table 7. Correlations above a moderate threshold level (Spearman's rank coefficient higher than 0.2) are underlined in the table.

By analyzing the trends in Fig. 8, we were first surprised to see that as a practitioner gets more years of work experience, against what

one would expect, s/he experiences more challenges in SDLC phases and SE tasks as listed; note that only the design phase of the SDLC is an exception to this observation. To explain this surprising observation, we explored potential root causes for these situations and what we found in talking to a few of our industry partners was that, as a practitioner gets more years of work experience, s/he usually takes a more active role in the project/teams, which usually means tackling large parts of a given project and thus facing more challenging issues.

Out of all the SDLC phases and SE tasks and support activities shown in Fig. 8, the five highest correlation values with increasing years of work experience are: communication with team members (Spearman's rho value = 0.29), project management (0.27), technical documentation (0.25), configuration management (0.23) and engineering process (0.21). This denotes the need for further investigations on finding the root causes of facing high challenges in these specific areas and, of course, need for more work by the research community in facilitating those challenges.

4.3.2. RQ 9: usage of SE tools versus years of work experience (Q42-2)

The goal of RQ 9 was to assess the extent to which a practitioner uses SE tools versus his/her years of work experience. It may be the case that, the more years of work experience a practitioner has, the more likely s/he will use SE tools in SE tasks, e.g., specific requirements engineering tools for the requirements analysis phase. In the online survey (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we had provided options for choosing the usage frequency of the following eight types of SE tools on a 5-point Likert scale: never (1), seldom (2), sometimes (3), frequently (4), and always (5):

- Software requirements tools (e.g., for documenting requirements)
- Software design tools (e.g., UML modeling tools)
- Software testing tools (e.g., for automating GUI testing)
- Software quality tools (e.g., for static code analysis)
- Software maintenance tools (e.g., for program comprehension or reverse engineering of design artifacts)

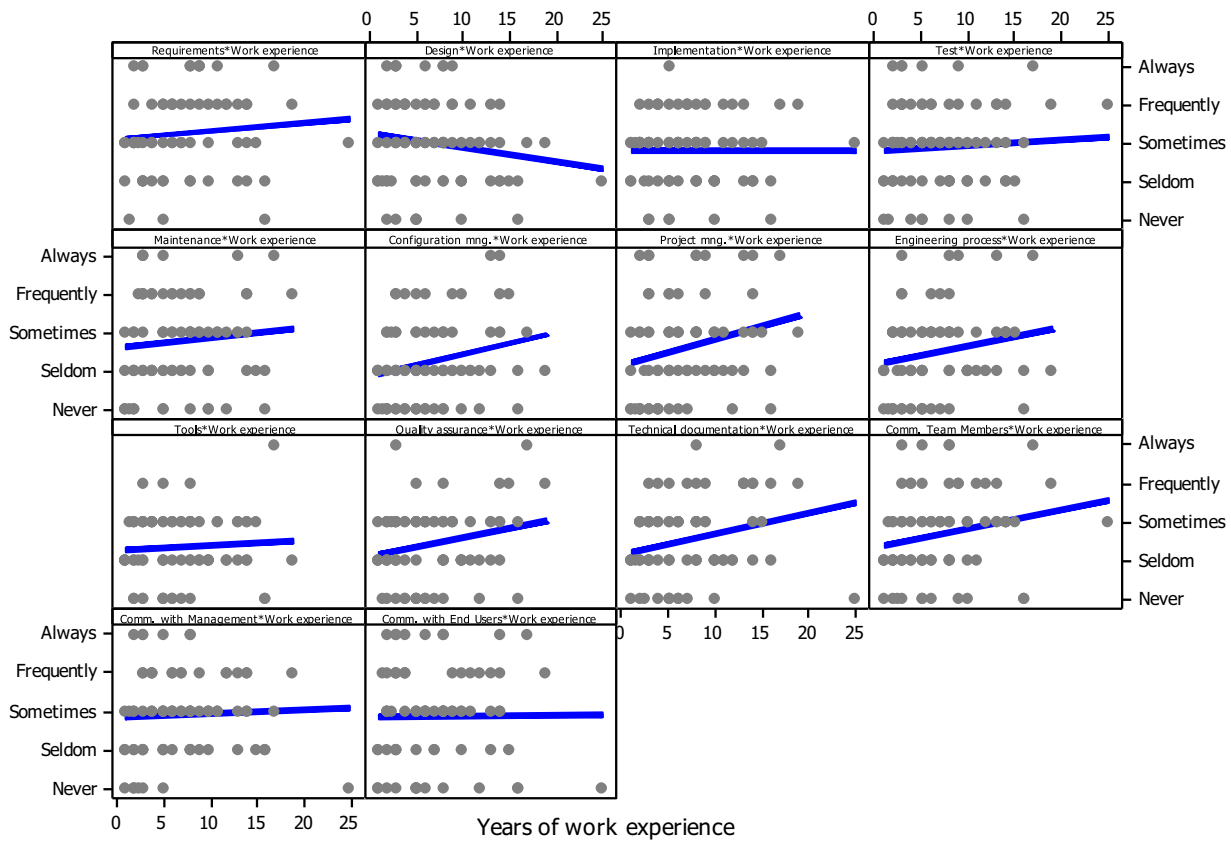


Fig. 8. Challenges experienced in SE phases/tasks (requirements, design, implementation, test, maintenance, configuration management, project management, engineering process, tools, quality assurance, technical documentation, communication with team members, communication with management, and communication with end users) versus years of work experience.

- Software engineering process tools (e.g., for process modeling and process management)
- Software configuration management tools (e.g., for version management, release and build)
- Software engineering management tools (e.g., for project planning and tracking, and risk management)

There was also a text box in the online survey for participants to put the name and type of any other SE tools they were using.

Fig. 9 shows, as an individual-value plot, the trends separated by the above eight types of SE tools. Correlation values are shown in Table 8. For five of the eight types of SE tools, we can see weak to moderate correlations, denoting that, for a typical practitioner, we can expect that with higher years of work experience, s/he would make more (frequent) use of SE tools. The top three highly-used SE tools in this context are: software quality tools, configuration management and project management tools. This denotes that, the more senior a software engineer, the more likely s/he will use those tools. Again, future studies are needed to probe into the reasons on high usage of those tools and less usage of other tools.

4.4. Cross-factor analyses among SE practices

RQs 10 and 11 are discussed in this section.

4.4.1. RQ 10: Challenges experienced in SDLC phases and SE tasks versus one another (Q14–14)

RQ 10 aimed at exploring the correlations among the degree of challenges experienced in various SE phases/tasks. For example, if a respondent has found requirements engineering highly challenging, s/he may also face challenges in software maintenance.

Similar to RQ 8, in our survey questionnaire (Garousi et al., 2014, Garousi et al., 2013, Garousi et al., 2015), we provided the list of five

SDLC phases and a set of nine SE (support) tasks, as shown in Table 9. Spearman's rank correlation coefficient values for correlations between levels of challenges experienced in different SDLC phases and SE tasks are provided in Table 9. A heat-map based color coding has been used as well to better highlight the trends. The five most correlated pairs of challenge areas are discussed below:

- Engineering process and SE tools (Spearman's rank correlation = 0.67): This is an interesting phenomenon since many participants found these two somewhat unrelated issues (process and SE tools) to be challenging together in their projects.
- Project management and engineering process (correlation = 0.62): These two issues are quite related since project management practices are usually applied along software processes. Thus, it is quite natural that participants, who face challenges in one of them, would also face challenges in the other too.
- Quality assurance and technical documentation (correlation = 0.61): This high correlation is also an interesting and somewhat unexpected phenomenon. However, as we can see, many participants who have one of these two tasks challenging, have also found the other challenging.
- Requirements and design (correlation = 0.60): These two issues are also related since, in the SDLC, design follows requirements and thus a challenging requirements phase could also denote a consecutive challenging design phase, e.g., the high complexity (or volatility) of system to be build could cause this.
- Software testing and maintenance (correlation = 0.60): These two issues are also related since, in the SDLC, maintenance follows the testing and delivery phases. Also, for some, testing is considered a type of maintenance activity, e.g., corrective maintenance.

Table 7
Correlations between the scales of challenges experienced in SE phases/tasks versus years of work experience.

	Requirements	Design	Implementation	Test	Maintenance	Configuration mng.	Project mng.	Engineering process	Tools	Quality assurance	Technical documentation	Comm. Team Members	Comm. with management	Comm. with end users
Spearman's rho	0.18	-0.11	-0.02	0.03	0.12	0.23	0.27	0.21	0.05	0.17	0.25	0.29	0.15	0.07
p-value	0.16	0.34	0.90	0.77	0.28	0.05	0.02	0.07	0.65	0.14	0.03	0.01	0.18	0.57

The four least correlated pairs of challenge areas are the listed below:

- Project management and implementation (Spearman's rank correlation = 0.03)
- Quality assurance and communication with management (correlation = 0.08)
- Technical documentation and implementation (correlation = 0.08)
- Communication with end users and configuration management (correlation = 0.08)

The two issues in each of the above five least correlated pairs of challenge areas are quite unrelated, thus facing challenges in one (e.g., quality assurance) will not imply facing challenges in the other (e.g., communication with management).

Furthermore, we observe in Table 9 that no pair of tasks has negative correlations with respect to challenges, i.e., high challenge in one task would mean low challenge in the other. In other words, if such a pair had existed, it would have meant that a high challenge level in one of the tasks would “ease” the job of practitioner in performing the other task and thus making it less challenging. Follow-up studies are needed to investigate the root-causes of the above observations.

4.4.2. RQ 11: using various software development methodologies by the same participant (Q15-15)

RQ 11 aimed at exploring extent to which different software development methodologies are used (or not used) by the same participant. For example, does wide adoption of the Waterfall methodology mean narrow adoption of Agile development?

Table 13 presents the Chi-square test values for usages of pairs of different software development methodologies and the corresponding *p*-values. Note that the Chi-square test was used since both variables are categorical scale and for all tests degrees of freedom (DF) is 1. Significant correlations with a *p*-value below 0.05 are underlined. We discuss the main observations in the followings:

- We notice that waterfall and Agile methodologies have significant correlation ($\chi^2 = 9.381$, *p*-value = 0.002). Odds ratio ((10/40) / (22/22) = 0.25) suggests that the odds of Agile usage is 4 times higher if a respondent is not using Waterfall than if using Waterfall, i.e., when practitioners are using waterfall, usage of Agile is 4 times less likely. The counts of responses for each pair of Waterfall/Agile combination usage are shown in Table 10.
- Extreme programming (XP) and Scrum have correlation (with chi-square values around 12.3 and 9.0, and *p*-values 0.000 and 0.003, respectively) with Agile/lean development, denoting that if the former practice is used, it is likely that the latter practice may also be used (in other projects inside the current employer companies the respondents). Based on the odds ratios for Agile/XP ((11/21) / (4/58) = 7.6) and Agile/Scrum ((12/20) / (7/55) = 4.7), odds of Extreme programming and Scrum usage is 7.6 and 4.7 times higher respectively if the practitioner is using Agile. These results are not surprising, as XP and Scrum are among the most popular Agile approaches (Dingsøyr et al., 2012). However, it is interesting that some practitioners using Agile/lean development did not report to use XP or Scrum. Moreover it is odd that some (although a minority) of the practitioners, who use XP or Scrum, did not report to use Agile/lean development. Perhaps this is due to a lack of knowledge of these terminologies. These findings would require further analysis for causes (is it because practitioners tend to use not all but some Agile approaches, or practitioners lack domain knowledge about Agile). The counts of responses for each pair of Agile/XP and Agile/Scrum combination usage are shown in Table 11.
- *P*-values in Table 13 also reveal that Extreme Programming might be correlated with both Scrum and Product-line development. However, we cannot conclude about correlations, since Chi-square test fails to satisfy one of its assumptions in both cases due

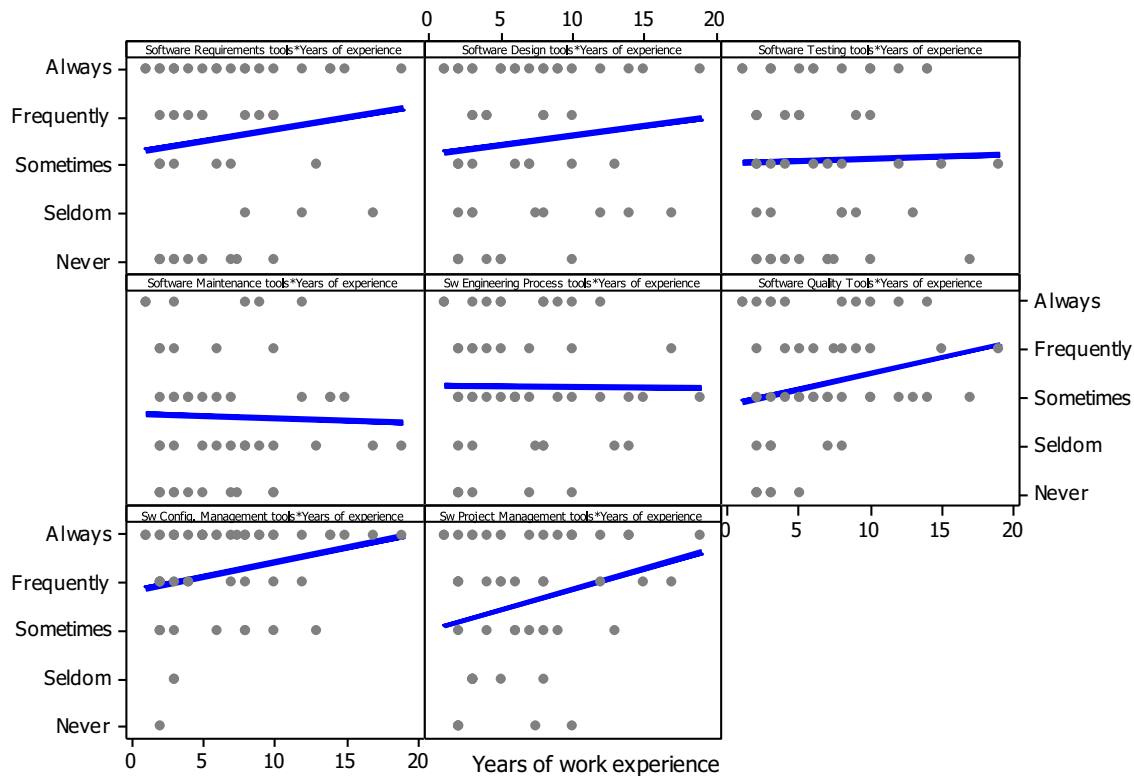


Fig. 9. Usage of SE tools (software requirements, design, testing, maintenance, engineering process, quality, configuration management, and project management tools) versus years of work experience.

Table 8
Correlations between the usage of SE tools versus years of work experience.

	Requirements tools	Design tools	Testing tools	Maintenance tools	Engineering process tools	Quality tools	Configuration management tools	Project management tools
Spearman's rho	0.14	0.13	0.05	−0.02	−0.01	<u>0.22</u>	<u>0.25</u>	<u>0.25</u>
p-value	0.34	0.37	0.76	0.90	0.98	0.14	0.08	0.08

to loss of statistical power as there are expected frequencies less than 5 (Field, 2013) as depicted in Table 12.

4.5. Cross-factor analyses among demographic aspects

The last question, RQ 12, is discussed in this section.

4.5.1. RQ 12: target industry sector versus company size (Q6-7)

We were interested in the relationships between target sector and company sizes. We would generally expect that larger companies are focusing on specific industries, e.g., defense. To assess RQ 12, the corresponding data have been visualized as a radar chart in Fig. 10.

A large portion of respondent companies which operate in the military and defense sector (around 90%) are larger companies (between 150 and 500 employees and more than 500 employees). For software companies active in the IT and telecommunication sectors, majority are smaller companies (less than 10 employees) with some medium size companies (less than 50 people). In the engineering/manufacturing software sector, most companies employing the respondents are still small size (6–10 people). Thus, we can notice that a large ratio of companies is small and medium-sized enterprises (SMEs), and would thus expect that the SE practices specific to SMEs are practiced in these companies. Follow-up studies are needed to investigate such SME-specific practices (similar to related work in this area, e.g., Paternoster et al. (2014)) and to see how and why SE practice in large companies versus SMEs differ.

4.6. Exploratory factor analysis (EFA)

To address RQs 1...12, we conducted multiple cross-factor analyses of pairs of factors in our study. As discussed in Section 1, the notion of cross-factor analysis as we conducted in this paper is different than Exploratory Factor Analysis (EFA) (Costello and Osborne, 2005, Child, 1990) and the wording similarity is coincidental. To conduct more rigorous statistical analysis, we decided to utilize EFA as well, as the last step in our study, to explore and uncover any underlying structure in our data set, not captured by our 12 RQs, to see if any other relationships hold among the data and items in our survey questions. We conducted two EFA analysis: for the group of development practices, and for the group of testing practices, as reported next.

4.6.1. EFA results for development practices

Principal component analysis with orthogonal rotation was applied on the 13 items within development related practices (Q24). Data was suitable for factor analysis with a Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy having a value of 0.751 (i.e. larger than 0.5) and Bartlett's test of sphericity significant at 0.00 (i.e. smaller than 0.05) (Field, 2013). As a result of an initial analysis, four components having eigenvalues above 1 (Field, 2013) were retained explaining 66.1% of the variance combined. After the final analysis, items were clustered under these four components. We used

Table 9
Correlations between challenges experienced in different phases.

	Req.	Design	Imp.	Test	Main.	Conf. management	Project management	Eng. process	Tools	Quality assurance	Tech. doc.	Comm. Team Members	Comm. with Mang.
Design	0.60												
Implementation	0.35	0.53											
Test	0.47	0.32	0.30										
Maintenance	0.32	0.37	0.14	0.60									
Configuration management	0.29	0.12	0.12	0.33	0.44								
Project management	0.34	0.15	0.03	0.36	0.41	0.40							
Engineering process	0.33	0.17	0.25	0.30	0.32	0.33	0.62						
Tools	0.16	0.19	0.29	0.24	0.31	0.33	0.42	0.67					
Quality assurance	0.28	0.11	0.15	0.46	0.43	0.49	0.57	0.54	0.48				
Technical documentation	0.34	0.24	0.08	0.34	0.30	0.39	0.45	0.39	0.31	0.61			
Comm. Team Members	0.39	0.18	0.15	0.24	0.41	0.16	0.41	0.25	0.28	0.21	0.34		
Comm. with Management	0.52	0.39	0.15	0.28	0.25	0.14	0.28	0.12	0.11	0.08	0.20	0.51	
Comm. with End Users	0.43	0.38	0.16	0.24	0.35	0.08	0.28	0.25	0.25	0.22	0.28	0.33	0.55

Table 10
Waterfall versus Agile/Lean practitioner counts.

		Agile / Lean		
		No	Yes	Total
Waterfall	No	22	22	44
	Yes	40	10	50
	Total	62	32	94

the EFA feature of the SPSS software. Table 14 presents the rotated factor loadings as an item-component matrix where only the factor loading values satisfying the minimum level for explanation (i.e. values above 0.4 (Field, 2013)) are shown. Note that the four group (cluster) labels under 'Rotated Factor Loadings' (the EFA terminology) in columns (e.g., quality-aware implementation and build practices) are the phrases that we have derived by looking at the similarities of the items clustered under them, as per the EFA results reported by the SPSS software.

The EFA grouped the development-related practices under four clusters as we discuss next:

- Quality-aware implementation: managing code complexity, code inspections, and built-in documentation,
- Build process: having a dedicated build person, a documented build process and pre-build code reviews,

- Developer responsibilities: conducting design and coding together and developers being responsible for design and building the product each day,
- Extreme programming concepts: pair programming and refactoring.

4.6.2. EFA results for testing practices

Another analysis was performed on the 14 items for testing related practices (Q25) and testing activity types (Q27) by using principal component analysis with orthogonal rotation. KMO measure with a value of 0.736 verified the sampling adequacy and Bartlett's test of sphericity significant at 0.00 suggested a sufficiently large correlation between items. Five components all having eigenvalues over 1 were retained. These five components cumulatively explains 79.8% of the total variance. Rotated factor loadings with a value over 0.4 are presented in Table 15. By using the rotated factor loadings, names of the components and items clustered under each component are presented below;

- Testing quality attributes includes load/stress testing, security testing and performance testing,
- Having a test team includes having a separate team for testing, developers and testers working together and independent testing by a test team,
- Unit testing covers developing and performing unit tests,
- Levels of testing includes user acceptance and integration testing,
- Roles of external stakeholders in testing is composed of only the participation of managers, clients or support people in testing.

Table 11
Extreme programming and Scrum versus Agile / Lean practitioner counts.

		Extreme programming					Scrum		
		No	Yes	Total			No	Yes	Total
Agile / Lean	No	58	4	62	Agile / Lean	No	55	7	62
	Yes	21	11	32		Yes	20	12	32
	Total	79	15	94		Total	75	19	94

Table 12
Scrum and product-line development versus extreme programming practitioner counts*, **.

		Scrum					Product-line development		
		No	Yes	Total			No	Yes	Total
Extreme programming	No	66	13	79	Extreme programming	No	73	6	79
	Yes	9	6*	15		Yes	10	5**	15
	Total	75	19	94		Total	83	11	94

* Cell with an expected frequency of 3.03.

** Cell with an expected frequency of 1.76.

Table 13
Correlations among the usage of software development methodologies.

	Waterfall	Prototyping	Incremental	Spiral	Agile / lean	Extreme programming	Scrum
Prototyping	0.006 (Chi-square test value) 0.937 (p-value)						
Incremental	0.835	3.677					
	0.361	0.055					
Spiral	0.000	0.287	1.707				
	0.990	0.592	0.191				
Agile/lean development	9.381	0.172	3.631	1.567			
	0.002	0.679	0.057	0.211			
Extreme Programming	2.827	0.523	0.186	1.149	12.271		
	0.093	0.469	0.666	0.284	0.000		
Scrum	1.175	3.494	2.997	0.001	8.99	4.333	
	0.278	0.062	0.083	0.982	0.003	0.037	
Product-line development	1.910	0.559	0.641	1.189	0.030	8.082	0.385
	0.167	0.455	0.423	0.275	0.863	0.004	0.535

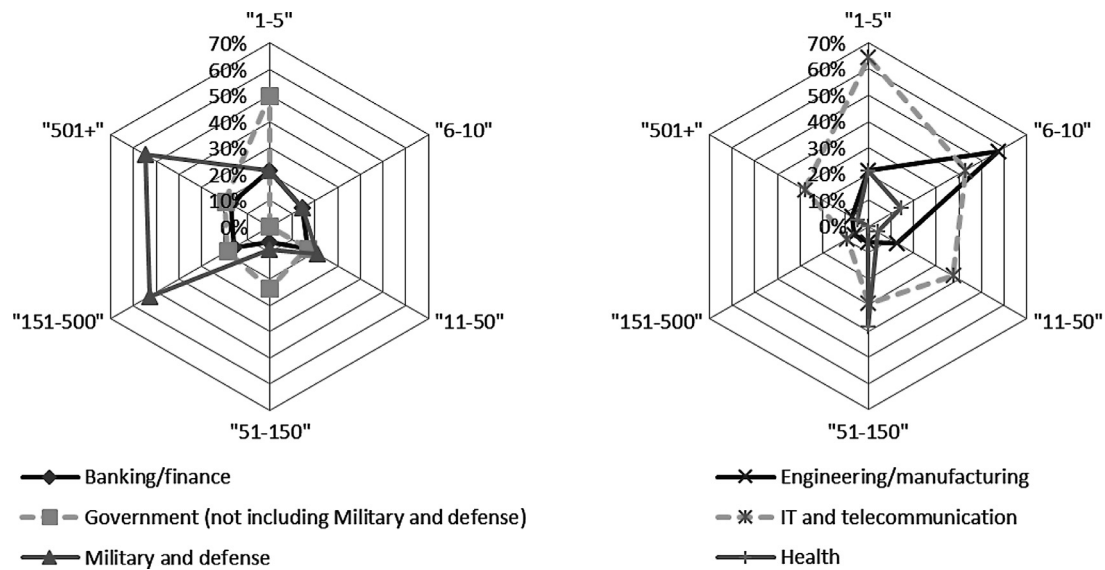


Fig. 10. Target industry sector versus company size.

Table 14

Exploratory factor analysis results for development practices (Q24).

Items	Rotated factor loadings			
	Quality-aware implementation	Build practices	Developer responsibilities	Extreme programming concepts
Code complexity	0.796			
Code inspections	0.771			
Built-in documentation	0.747			
Static code analysis	0.576		0.517	
Dedicated build person		0.836		
Build process documented		0.757		
Code reviewed before build		0.703		
Dev. responsible for design			0.776	
Product built each day			0.718	
Build after a feature		0.500	0.634	
Design and coding together			0.604	
Pair programming				0.805
Refactoring				0.789
Eigenvalues	4.202	2.006	1.300	1.086
% of variance	32.322	15.429	9.997	8.352

Table 15

Exploratory factor analysis results for testing practices and activity types (Q25 and Q27).

Items	Rotated factor loadings				
	Testing quality attributes	Having a test team	Unit testing	Levels of testing	Roles of external stakeholders in testing
Load/stress testing	.849				
Security testing	.831				
Performance testing	.806				
Unit tests formally reviewed	.631		.543		
Separate team for testing		.903			
Developers and testers work together		.837			
All new features independently tested by a test team		.835			
Unit testing			.894		
Develop unit tests			.870		
User acceptance testing				.873	
Integration testing				.624	
Functional/system testing		.515		.534	
A manager, client advocate or support person helps testing					.839
Developer tests the product			.455		.674
Eigenvalues	5.435	2.046	1.380	1.205	1.107
% of variance	38.824	14.616	9.854	8.607	7.905

5. Discussions

We summarize in the following the findings of this study and then discuss the lessons learned, limitations, potential threats to validity, and the steps that we have taken to minimize or mitigate them.

5.1. Summary of findings

Summary and implications of findings from each RQ are as follows.

RQ 1 (use of size measurement approaches versus target sector type):

- The LOC metric is a very popular size measure in military defense sector with a usage rate of 73% which is perhaps due to the higher maturity levels of these companies employing the respondents in general.
- Point-based estimation metrics (i.e., COSMIC FP, IFPUG FP, use-case points and story points) are used rarely.

RQ 2 (usage of software development methodologies versus target industry sector):

- Agile/lean development is used the least (16%) by the participants employed by the companies serving the military and defense sector. There have been similar observations in this regard as well, e.g., necessary considerations and concerns for the US Department of Defense (DoD)'s Agile adoption have been discussed in [Lapham \(2012\)](#) which reported low adoption of Agile in the DoD.
- Agile and lean development is the most popular methodology among participants developing software for engineering/manufacturing, IT and telecommunication and health sectors.
- Scrum, spiral and product-line development are not very popular in any of the target sectors.

RQ 3 (usage of software development methodologies versus company size):

- Usage of waterfall is low among practitioners employed by small-sized companies whereas Agile/lean development is relatively popular among this class of practitioners.
- As company size increases, usage of spiral development slightly increases, whereas usage of extreme programming practices decreases.
- We inferred that participants employed by larger companies still favor waterfall-like development whereas using Agile/lean is preferred more in smaller companies.

RQ 4 (importance of design-related quality attributes versus target industry sector):

- For respondents employed by companies targeting engineering/manufacturing industry sectors, usability considerations seem to be very important.
- For all the current employer companies of the participants, regardless of their target industry sectors, different design-related quality attributes are “sometimes” to “always” important.

RQ 5 (test types/levels versus company size):

- As company size grows, there is more chance for integration testing to be conducted.
- There are noticeable correlations (Spearman's rank correlation values > 0.5) among the different types of non-functional testing (i.e., load, stress, security and performance testing), i.e., if a practitioner decides to conduct one

of these types of testing, it would also likely conduct the other non-functional testing types as well.

RQ 6 (types of software maintenance activities versus number of projects):

- We notice that for the cases of adaptive and perfective maintenance, there are weak positive correlations with increase in number of projects, i.e., if a company has more projects at the same time, practitioners will conduct more software maintenance activities of the above types.

RQ 7 (usage of SE tools versus number of projects):

- The higher the number of projects, the more a practitioner would use SE tools.

RQ 8 (challenges experienced in SE phases/tasks versus years of work experience):

- It was surprising to see that as a practitioner gets more years of work experience, s/he experiences more challenges in SDLC phases and SE tasks. By talking to a few of our industry partners, we found out that, as expected, as a practitioner gets more years of work experience, s/he usually takes a more active role in projects/teams, which usually means tackling large parts of a given project and facing more challenging issues in SDLC phases and SE tasks.

RQ 9 (usage of SE tools versus years of work experience):

- For a typical practitioner, by higher years of work experience, s/he would make more (frequent) use of SE tools.

RQ 10 (challenges experienced in SDLC phases and SE tasks versus one another):

- The top most correlated pairs of challenge areas were: (1) engineering process and SE tools, (2) project management and engineering process, (3) quality assurance and technical documentation, (4) requirements and design (i.e., facing challenges in requirements would also mean facing challenges in design), and (5) testing and maintenance.

RQ 11 (using various software development methodologies by the same practitioner):

- Waterfall and Agile methodologies had negative correlation, denoting that if one is used by a practitioner, the other will less likely be used.

RQ 12 (target industry sector versus company size):

- A large portion of current employer companies of the respondents, which operate in the military and defense sector (around 90%), are larger companies (between 150 and 500 employees and more than 500 employees).
- For software companies active in the IT and telecommunication sectors, majority are smaller companies (less than 10 employees) with some medium size companies (less than 50 people).

5.2. Lessons learned

During this study, we have learned a number of lessons especially in the area of survey design and strategy. Our experience throughout this study revealed once again the importance of proper survey design and strategy. Authors' experiences in previous survey studies ([Geras et al., 2004](#), [Garousi and Varma, 2010](#)) were helpful in both

design and execution of the survey in the current paper. We categorized the questions based on the SWEBOK guideline (version 2004) (Bourque and Fairley, 2004). Also in design of the survey questions, we not only used our domain knowledge but also benefited from reviewing similar surveys and consulting industrial partners. These efforts aimed at ensuring use of a terminology that is familiar to the respondents and keeping the questions at a reasonable number without sacrificing domain coverage. In execution, we devised a progressive publicity plan. Each publicity task was initialized with a few days apart. This enabled us to monitor and control the survey population as we were able to observe exponential increases in total number of participants after each invitation sent.

In this study, we have benefitted from defining RQs in establishing research directions. We used RQs in selecting the cross-factor combinations that were found to be the most meaningful and important in the context of latest software engineering practices and trends. Selection of those most important combinations required brainstorming among authors and also consultations with several key industry contacts. RQs were helpful not only in setting the scope of the study (i.e. selecting the combinations) but also in analyzing survey data towards our intended purposes. We relied on our pre-established RQs for grouping and classifying the analysis results, choosing the perspectives that data analysis is based on and deciding on the graphical and tabular representations that would serve best to answer the questions and reach our intended goals.

We have reviewed and reported a large number of related work in Section 2. However, studies that report cross-factor correlations as we do in this study are not many. Since we did not find any existing work to analyze the cross-factor combinations that we have considered, we had no opportunity to compare and contrast our findings with other studies. We would have benefited from raw data in creating cross-factor analysis to be compared in that sense. Therefore, realizing the importance of open raw data for replication and use in future studies, we make our survey data publicly available online (Garousi, Oct. 2014) for SE practitioner and researchers.

5.3. Limitations and potential threats to validity

We discuss in the following, the possible validity concerns, the resulting limitations of our study in terms of internal, construct, conclusion and external validity concerns, and also the steps that we have taken to minimize or mitigate them.

5.3.1. Internal validity

Internal validity is a property of scientific studies which reflects the extent to which a causal conclusion based on a study is warranted (Wohlin et al., 2000). A threat to internal validity in this study lies in the randomness of the participants. We tried to achieve a random set of samples by setting a variety of publicity tools in order to reach out to as many participants and also various demographics of the Turkish software engineer population as possible.

5.3.2. Construct validity

Construct validities are concerned with issues that to what extent the object of study truly represents theory behind the study (Wohlin et al., 2000). What we did was common to other survey studies—we counted the votes for each question and then made statistical inferences. It is believed that results based on such voting data can, to a certain extent, reflect the opinions of the majority of Turkish practitioners.

It is also common for people to deflect their answers when they feel being evaluated and based on what they think is the intended result of a study. To mitigate these, we informed participants prior to the survey that our motive in this study was to take a snapshot of the industry and that we will not collect any identifying information so that participants will remain anonymous.

5.3.3. Conclusion validity

Conclusion validity of a study deals with whether correct conclusions are reached through rigorous and repeatable treatment (Wohlin et al., 2000). We attempted to conclude, qualitatively, that the SE practices have economic and psychological aspects as well as purely technical concerns. For each RQ, we attempted to reduce the bias by seeking support from the statistical results. Thus, all the conclusions that we drew in this study are strictly traceable to data. Moreover, to increase transparency, the raw survey data is made available online for other researchers to validate and replicate.

Our survey questions were reviewed by the authors of this study and a few industry contacts for preventing survey reliability related issues such as poor question wording and survey layout. Furthermore, we improved the reliability of our survey (treatment) using a small-scale pilot study prior to the main survey execution.

5.3.4. External validity

External validity is concerned with to what extent the results of this tertiary study can be generalized (Wohlin et al., 2000). Our participants were mainly invited through the researchers' network of partners/contracts in Turkish software industry. Companies out of our contact network were not probably properly represented in the survey population. To mitigate this validity issue, we took the following steps:

- We attempted to enlarge our email-list to include all major Turkish companies and technology development zones
- We posted messages in online Social and professional networks, e.g., LinkedIn, Yahoo Groups and Twitter
- We allocated two months of time for all invitees to fill out the online questionnaire.

Our sample size and geographic distribution of samples are quite reasonable in the context of Turkish. However, clearly, we do not have any intentions to generalize our findings to other countries and regions. Nevertheless, we reported demographic information of the participants, projects and companies covered in our study, and therefore the readers will be able to evaluate the applicability in different contexts.

6. Conclusions and future work

We devised this study to understand how software engineering practices are related to practitioner demographics in Turkey. Software engineering researchers and practitioners in the field have different beliefs on the cross-factor relations that we investigated in this study, and they rarely consider statistical evaluations for those factors. We used data from a recent survey on software engineering practices to shed light on these relations in the context of the Turkish industry.

It is interesting to see how practitioners employed by software companies operating in the military and defense sector differ from others in Turkey. Our findings shows that practitioners employed by military and defense sector companies as significantly larger companies (RQ 12) differ themselves in some practices such as usage of size measures (RQ 1) and Agile/lean development methods (RQ 2). Causes of this might be related to a handful of reasons ranging from development traditions in the sector to process maturity model adoption habits to organizational structure of these large companies.

It is worth mentioning that some of the findings were as one would expect, e.g., if a practitioner uses one of the Waterfall or Agile/Lean methodologies, s/he less likely uses the other one (RQ 11). Moreover, not surprisingly Waterfall is still more popular in larger companies whereas practitioners employed by smaller companies favor Agile more (RQ 3). Also another such expected finding was that practitioners in larger companies, who expectedly develop more

complex products, are more likely to perform integration testing (RQ 5). It is also not surprising to find out that software engineers, who face challenges about requirements, also have difficulties about design and vice versa (RQ 10).

It is discouraging to see that some trending software engineering practices and methods are not very popular in Turkey. These include the usage of point-based estimation metrics, Scrum and product-line development for practitioners employed by software companies from almost all target sectors.

We plan to investigate the following future work directions:

- Further similar studies are needed in other regions and countries to be able to compare the latest trends in software engineering practices.
- We explored the usage of SE tools versus number of projects in RQ 7. However, probing and exploring the success stories and challenging experiences in using SE tools in the context of Turkish software industry would be an interesting investigation in future studies.
- Challenges experienced in SDLC phases and SE tasks and their cross-factor relationships were explored by RQ 10. Follow-up studies are needed to investigate the root-causes of those challenges.

Acknowledgments

We would like to sincerely thank all the software engineers from across Turkey who anonymously participated in the survey. Last but not least, we are thankful to our colleagues (Aysu Betin-Can, Deniz Akdur, Özden Özcan Top, Elif Aydın, Burak Coşkun, and Ergin Topçu and others) who helped in the publicity of our survey. Vahid Garousi was partially supported by several internal grants provided by the Hacettepe University and the Scientific and Technological Research Council of Turkey (TÜBİTAK).

Appendix. Survey questions

The complete list of the questions used in the survey is shown in Table 16. For brevity, we are not showing the pre-designed multiple-choice answers of the questions in Table 16, but they can be found in an online resource (Coşkunçay et al., 2013) and the two related studies (Garousi et al., 2013, Garousi et al., 2015). Several questions (e.g., #14) had answers of type 5-point Likert scale. For those questions, the five choices were: never (1), seldom (2), sometimes (3), frequently (4), and always (5).

Table 16
Questions developed and used in the survey.

RQ	Aspect	Survey questions (and Metrics)	Type of answers					
			Single answer	Multiple answers could be chosen	Likert scale	Integer	Free text field	Binary answer (yes or no)
1	Profiles and demographics of practitioners including their companies and projects	Practitioners:						
		1. What is (are) your current position(s)?		x			x	
		2. How many years of work experience do you have in IT and software development industries?				x		
		3. What is your highest academic degree?	x					
		4. What is (are) your university degree(s) in?		x			x	
		5. Please choose the city that you work in?		x			x	
		Companies:						
		6. What is the target sector of the products developed by your company?		x			x	
		7. What is the size of your company (number of employees)?		x				
		Projects:						
		8. What kind of software do you develop?		x				
		9. How do you classify the software packages that you develop in your company?		x			x	
		10. In case you have measured the size measure in your latest development project, please indicate the methodology used (COSMIC Function Points, IFPUG Function Points, Use-Case Points, Story Points, Lines of Code, etc.) and the approximate size of your recent project?		x			x	
2	Overall characteristics of the Software Development Life Cycle (SDLC) and development processes	11. What is the number of software projects undertaken in your company the past two years?		x				
		12. How much effort in % is spent by your team in the past projects on different SDLC phases?				x		
		13. How much are you involved in different SDLC phases?			x			

(continued on next page)

Table 16 (continued)

RQ	Aspect	Survey questions (and Metrics)	Type of answers					
			Single answer	Multiple answers could be chosen	Likert scale	Integer	Free text field	Binary answer (yes or no)
3	Software requirement	14. How much challenge do you experience in each of the SDLC phases/tasks?			x			
		15. What kind(s) of software development methodologies do you use in your team?		x			x	
		16. Do you formally follow any of the well-known process improvement models?		x			x	
		17. How often are various SE process practices carried out by your organization?			x			
		18. How often are various software requirement practices carried out by your organization?			x			
4	Software design	19. What type of notation does your team use to document software requirements?		x			x	
		20. How often are various software design practices carried out by your organization?			x			
		21. What types of software design activities do you perform?			x			
		22. How important is each of the software quality factors when you design a software system?			x			
5	Software development	23. Which programming languages do you use in your company?		x			x	
		24. How often are various software development practices carried out by your organization?			x			
6	Software testing	25. How often are various software testing practices carried out by your organization?			x			
		26. In terms of the type and phasing of testing during the SDLC, what is the type of your test activities?			x		x	
		27. How often do you conduct the different types of test activities?			x		x	
		28. In your current or most recent software project, what test technique did the team use to generate test cases?		x			x	
		29. Overall in all of your past projects, how much automated versus manual testing have you done?				x		
		30. How often do you use various code (test) coverage metrics in your test activities?			x		x	
		31. How often do you use various other test and quality metrics do you explicitly measure in your projects?			x		x	
		32. Test management: Please identify the ratio of testers to developers (testers: developers) in your current or most recent project. For example, if you had on average one tester for every 2 developers, the answer would be 1:2.		x				
		33. What criteria are used in your projects to decide that the testing activities are terminated/completed?		x			x	
		34. How often are various software maintenance practices carried out by your organization?			x			
7	Software maintenance	35. What is the level of challenge you experience in various software maintenance practices?			x			
		36. If you have reported challenges in the above question, please summarize the challenges.					x	

(continued on next page)

Table 16 (continued)

RQ	Aspect	Survey questions (and Metrics)	Type of answers					
			Single answer	Multiple answers could be chosen	Likert scale	Integer	Free text field	Binary answer (yes or no)
8	Software configuration management, release planning and support practices	37. What types of challenges have you noticed in software maintenance tasks?		x			x	
		38. How often are various software release and delivery practices carried out by your organization?			x			
		39. What types of practices are carried out by your organization while supporting your software projects?						x
9	Software project management	40. How often are various software project management practices carried out by your organization?			x			
		41. What types of software project management practices are carried out by your organization?			x			
10	Usage of SE tools	42. Which tools are used for different software engineering tasks?		x			x	
11	Software quality assurance	43. What types of software quality assurance practices are carried out by your organization?			x			
12	Research activities and Interaction with Academia	44. Does your company have a dedicated R&D department/unit, specializing on SE research and aiming at developing new better ways/techniques to develop software?		x				
		45. How often do you interact (talk) to SE university researchers about your problems/challenges.		x				
		46. How often do you read technical papers (articles) published in SE journals, conferences or workshops?		x				

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